UNIVERSITI TEKNOLOGI MARA

THE IMPLEMENTATION OF THREE-TERM CONJUGATE GRADIENT METHOD IN INTENSITY-MODULATED RADIATION THERAPY

A'ISYAH KAMILAH BINTI MD HASSAN

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A'ISYAH KAMILAH BINTI MD HASSAN

Report submitted in fulfilment
of the requirements for the degree of
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APPROVED BY:



……..

DR. NORHASLINDA BINTI ZULL PAKKAL

Supervisor

Faculty of Computer and Mathematical Sciences

Universiti Teknologi MARA

Terengganu Branch

Kuala Terengganu Campus

DECLARATION BY THE CANDIDATE

I certify that this report and project to which it refers is the product of my own work and that any idea or quotation from the work of other people, published or otherwise, are fully acknowledged in accordance with standard referring practices of this discipline.



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A'ISYAH KAMILAH BINTI MD HASSAN

Faculty of Computer and Mathematical Sciences
Universiti Teknologi MARA
Terengganu Branch
Kuala Terengganu Campus

ABSTRACT

Optimization involves finding the optimal solution of an objective function. The Conjugate Gradient (CG) method is one of the most widely used methods for solving unconstrained optimization problems. It has been applied in various fields such as robotic motion, image restoration, and regression analysis. The CG method can be classified into several categories, including scaled, three-term, and hybrid CG methods. However, the performance of CG methods depends on the choice of the CG coefficient, and studies on the comparison of Three-Term Conjugate Gradient (TTCG) coefficients for large-scale inverse planning problems are still limited. This research studies the performance of TTCG methods under exact line search and their use in Intensity-Modulated Radiation Therapy (IMRT) fluence map optimization. Five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, are tested on 15 standard test functions with different initial points and variables from 2 to 10,000 variables. The performance is measured by the number of iterations (NOI) and central processing unit (CPU) time. The results are presented using performance profiles, where efficiency is measured by lesser NOI and CPU time, while robustness is measured by the ability to solve all test problems successfully. The TTLAMR coefficient achieves a 100% success rate and requires fewer iterations and lesser CPU time compared to most of the other TTCG methods. All five coefficients are then applied to a clinical prostate IMRT fluence map optimization problem. On this application, TTKMAR converges in the fewest iterations, while TTLAMR attains the lowest final objective value, and the leading coefficients all produce clinically acceptable treatment plans. This shows that the best coefficient on the standard test functions is not necessarily the best in the clinical application.

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LIST OF SYMBOLS

Symbols

x	Vector of decision variables
$f(x)$	Objective function to be minimized
g_k	Gradient vector at iteration k
d_k	Search direction at iteration k
α_k	Step size (step length) at iteration k
β_k	Conjugate gradient coefficient at iteration k
x_k	Approximate solution at iteration k
s_k	Difference between successive iterates, $x_{k+1} - x_k$
y_k	Difference between successive gradients, $g_{k+1} - g_k$
∇f	Gradient operator applied to f
$\| \ \|$	Euclidean norm
T	Transpose of a vector or matrix

LIST OF ABBREVIATIONS

Abbreviations

CG	Conjugate Gradient
TTCG	Three-Term Conjugate Gradient
SD	Steepest Descent
NOI	Number of Iterations
CPU	Central Processing Unit
IMRT	Intensity-Modulated Radiation Therapy
OAR	Organ at Risk
PTV	Planning Target Volume
DVH	Dose-Volume Histogram
CORT	Common Optimization for Radiation Therapy
TTRMIL	Three-Term Rivaie-Mustafa-Ismail-Leong coefficient
TTLAMR	Three-Term Linda-Aini-Mustafa-Rivaie coefficient
TTKMAR	Three-Term Kamilu-Mustafa-Abdelrahman-Rivaie coefficient
TTSMAR	Three-Term Sulaiman-Mustafa-Abdelrahman-Rivaie coefficient
TTSFA	Three-Term Saleh-Fadhilah-Aidya coefficient

LIST OF NOMENCLATURE

Nomenclatures

Beamlet	A small subdivision of a radiation beam whose intensity is optimized individually
Fluence map	The spatial distribution of beamlet intensities for a treatment beam
Inverse planning	A process that determines beamlet intensities from prescribed dose objectives
Exact line search	A line search that exactly minimizes the objective along the search direction
Sufficient descent	A condition ensuring the search direction produces a decrease in the objective
Global convergence	Convergence of an iterative method to a stationary point from any starting point
Performance profile	A graphical tool for comparing the efficiency and robustness of solvers
Gray (Gy)	The SI unit of absorbed radiation dose
D95	The dose received by 95% of the planning target volume
Conjugate direction	A search direction that is conjugate to previous directions with respect to a matrix
Ill-conditioned problem	A problem highly sensitive to small changes, causing slow or unstable convergence
Convergence	The process by which iterates approach an optimal or stationary solution

CHAPTER 1
INTRODUCTION

1.1 Introduction

Chapter one introduces the foundation of this research that includes the research background, motivation for this work, the problem statements, the research objectives and questions. It also highlights the significance and scope of the research, discusses the project benefits and provides definitions of terms and concepts related to the research. Lastly, this chapter presents an overview of the structure and the research outline of the entire research.

1.2 Research Background

Optimization plays an important role in various fields of science, engineering, economics, and medicine (Ibrahim et al., 2026). The objective of optimization is to determine the best solution among a set of feasible alternatives while satisfying specific requirements. Many real-world problems can be formulated as optimization problems, where the goal is to minimize or maximize an objective function (Ben Hanachi et al., 2022). Depending on the existence of constraints, optimization problems can generally be classified into constrained and unconstrained optimization problems. Among these, unconstrained optimization is given much attention because many large-scale applications and numerical algorithms are built on it.

The Conjugate Gradient (CG) method is one of the most common ways to solve unconstrained optimization problems. It was first introduced by Hestenes and Stiefel (1952) for solving linear systems and was later extended by Fletcher and Reeves (1964) to nonlinear unconstrained optimization problems. CG methods are popular because the computation is simple, the memory needed is small, and large-scale problems can be handled well. Unlike Newton-type methods, second derivatives are not computed or stored, so the cost stays low even when the problem has many variables (Ben Hanachi et al., 2022; Najmm & Ahmed, 2022).

The general iterative scheme of the CG method generates a sequence of approximations that progressively approach the optimal solution. The efficiency of the

method is largely influenced by the choice of the CG coefficient, which determines the search direction at each iteration. Consequently, numerous CG variants have been developed over the years, including the Fletcher-Reeves (FR), Polak-Ribiére-Polyak (PRP), Hestenes-Stiefel (HS), Dai-Yuan (DY), and Dai-Liao (DL) methods. These methods differ in the formulation of their CG coefficients and exhibit different convergence and computational characteristics (Alhawarat et al., 2024; Ben Hanachi et al., 2022).

Despite their advantages, classical CG methods may experience limitations such as slow convergence, poor descent properties, or reduced numerical efficiency when applied to certain optimization problems. As a result, researchers continue to develop modified, hybrid, and three-term conjugate gradient (TTCG) methods to improve the sufficient descent property, global convergence, and computational performance. Recent studies also have proposed various improvements to existing CG formulas and demonstrated their effectiveness through extensive numerical experiments on large-scale unconstrained optimization problems (Alhawarat et al., 2024; Ben Hanachi et al., 2022).

The effectiveness of CG methods has led to their successful application in numerous real-world problems. In machine learning, CG methods have been employed for neural network training and regression analysis. In image processing, CG-based algorithms have been applied to image restoration, image reconstruction, and image classification tasks. Furthermore, CG methods have been utilized in electrical engineering, power quality enhancement, discriminant analysis, and large-scale data analysis problems. These applications demonstrate the flexibility and robustness of CG methods in solving complex optimization problems across field of study (Alhawarat et al., 2024; Sahoo et al., 2023; Shi & Chen, 2024).

Among the many applications of optimization techniques, Intensity-Modulated Radiation Therapy (IMRT) has emerged as an important application in the medical field (Moreno et al., 2025). IMRT is an advanced radiation therapy technique used for cancer treatment, where radiation beams are modulated to deliver prescribed doses to tumour tissues while minimizing radiation exposure to surrounding healthy organs. The quality of an IMRT treatment plan depends heavily on the optimization process used to determine the beamlet intensity values. This process is commonly referred to as inverse treatment planning and is often formulated as a large-scale

optimization problem involving thousands of decision variables (Moreno et al., 2025; Obata et al., 2024).

The specific optimization problem solved in IMRT inverse planning is called fluence map optimization. In this process, each radiation beam is divided into many small beamlets, and the goal is to find the best intensity for each beamlet so that the delivered dose meets the clinical treatment objectives as closely as possible (Maass et al., 2022). The relationship between the beamlet intensities and the delivered dose is represented by a dose-influence matrix. The treatment goals are then written as an objective function, which aims to deliver the prescribed dose to the tumour while reducing the dose to the surrounding organs at risk (OAR). By minimizing this objective function, the optimal fluence map is obtained, which defines the treatment plan. Therefore, fluence map optimization is a key step in the IMRT optimization process.

The optimization problem in IMRT is computationally challenging due to the large number of beamlets and the complexity of the dose influence matrix (Barkmann et al., 2023). Consequently, efficient optimization algorithms are required to obtain high-quality treatment plans within a reasonable computational time. Since CG methods are designed for large-scale optimization problems and require relatively low memory storage, CG methods are well suited for solving IMRT inverse planning problems (Bamigbola, 2024; Ben Hanachi et al., 2022). Among the different CG methods, TTCG methods modify the search direction by using three terms instead of the conventional two-term formulation. The additional term is intended to improve the robustness and efficiency of the algorithm, making TTCG methods suitable for solving large-scale optimization problems.

Based on these advantages and the heavy computation in IMRT, this study implements a TTCG method in IMRT inverse planning. The method is first tested on standard unconstrained test functions and then applied to an IMRT treatment planning problem. Its efficiency is measured by the number of iterations (NOI) and central processing unit (CPU) time, which are the usual performance indicators in CG research (Mellal et al., 2024).

1.3 Motivation for This Work

The gradient-based optimization methods, particularly the CG method, are widely used for solving large-scale unconstrained optimization problems because of the

low memory requirement and the use of only first-order derivative information(Hager et al., 2006; Nocedal & Wright, 2006). These properties make CG methods suitable for solving IMRT inverse planning problems, where thousands of beamlet intensities must be optimized simultaneously. However, the classical two-term CG method may become less efficient for ill-conditioned problems, which are common in IMRT because of the complex dose influence matrix (Barkmann et al., 2023; Maass et al., 2022).

To overcome this limitation, many researchers have proposed TTCG methods. By introducing an additional term into the search direction, TTCG methods can improve the descent property and often provide better numerical performance than the classical two-term CG method (Zhang et al., 2004). Several TTCG methods, including TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, have been proposed in the literature. However, the performance of these TTCG methods has not been systematically compared for IMRT inverse planning.

Another important factor that affects the performance of CG methods is the line search strategy. A line search determines the step length at each iteration, and an appropriate step length helps improve the efficiency and convergence of the optimization algorithm (Nocedal & Wright, 2006). Among the available line search techniques, the exact line search computes the optimal step length along the search direction at every iteration. Although it requires more computation, it provides a fair and consistent basis for comparing different TTCG methods because the results are not affected by step-length approximation errors.

This study is therefore motivated by the need to identify an efficient TTCG method for IMRT inverse planning. The selected TTCG methods are first evaluated on standard unconstrained optimization problems and then implemented in an IMRT treatment planning problem. Their performance is compared using the NOI and CPU time to determine the most efficient TTCG method for solving the IMRT optimization problem.

1.4 Problem Statement

Optimization in IMRT is an important research topic because the radiation dose must reach the tumour precisely while the nearby OAR and healthy tissues are protected (Moreno et al., 2025). Thus, inverse optimization in IMRT treatment planning is used to determine the intensities of many beamlets based on prescribed dose

requirements. However, solving this problem is difficult because hundreds to thousands of beamlet intensities must be optimized at the same time, which leads to high computational cost (Broderick et al., 2009; Cho, 2018).

Several optimization methods have been proposed for IMRT inverse planning. For large-scale unconstrained problems like IMRT, the CG method is widely used because it needs little memory and only first derivatives, so thousands of variables can be handled efficiently (Abdelhamid et al., 2025; Alhawarat et al., 2021). To improve the performance of the classical CG method, TTCG methods have been developed by introducing an additional term into the search direction. The additional term is intended to improve the efficiency and robustness of the optimization algorithm. However, the performance of a TTCG method still depends on the choice of the CG coefficient, the line search strategy, and the search direction formula (Malik et al., 2023; Rivaie et al., 2012). Different TTCG methods may therefore require different NOI and CPU times to reach convergence. Although several TTCG methods have been proposed in the literature, their performance has not been systematically compared for IMRT fluence map optimization.

This study evaluates the performance of five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA. First, the TTCG methods are tested on standard unconstrained optimization test functions. The numerical performance of the TTCG methods is then compared in terms of the NOI, CPU time, and performance profiles to identify the most efficient method for solving standard optimization problems. Next, all five TTCG methods are implemented in an IMRT fluence map optimization problem. The performance of the TTCG methods is further compared in terms of NOI, CPU time, final objective value, and the percentage of rectum volume receiving more than 30 Gy. The findings from both the standard optimization test functions and the IMRT application are then used to determine the most suitable TTCG method for IMRT inverse planning. Therefore, this research is conducted to answer the following research questions:

i) How does the TTCG method perform when solving standard optimization test functions?

ii) How does the performance of the TTCG method vary in terms of NOI and CPU time?

iii) How do the TTCG methods perform when applied to an IMRT fluence map optimization problem?

1.5 Research Objectives

In accordance with the problem statement, the main objectives are:

i) To solve standard optimization test functions using TTCG methods under exact line search.

ii) To compute and analyze the performance of the TTCG methods in terms of NOI and CPU time.

iii) To implement the TTCG methods in an IMRT fluence map optimization problem.

1.6 Significance of Study

The CG method is widely used for solving large-scale unconstrained optimization problems because of its low memory requirement and computational efficiency. These advantages make the CG method suitable for applications such as IMRT, where many beamlet intensities must be optimized simultaneously. However, the computational performance of the CG method depends on the choice of the CG coefficient, the search direction, and the line search strategy. Therefore, selecting a suitable TTCG method is important for improving the efficiency of large-scale optimization problems.

This study contributes to the field of numerical optimization by providing a systematic comparison of five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA. The comparison is carried out using the number of iterations NOI, CPU time, and performance profiles to identify the most efficient TTCG method for solving standard unconstrained optimization problems. The findings provide useful information for researchers in selecting an appropriate TTCG method for similar large-scale optimization problems.

This study also contributes to the application of optimization methods in IMRT. All five TTCG methods are implemented in an IMRT fluence map optimization problem and compared in terms of NOI, CPU time, final objective value, and the

percentage of rectum volume receiving more than 30 Gy. The findings provide a better understanding of the computational performance of different TTCG methods in IMRT inverse planning. An efficient TTCG method can reduce computational time while maintaining treatment quality, making IMRT treatment planning more efficient.

Overall, this study provides a useful reference for future research on TTCG methods and their application to large-scale optimization problems. The findings may also support the development of more efficient gradient-based optimization methods for IMRT and other real-world optimization applications.

1.7 Scope of the Project

In this project, the implementation and comparison of the CG method with the various CG coefficients, namely, TTRMIL, TTLAMR, TTKMAR, TTSMAR and TTSFA are considered. The primary goal of this research is to compare the computing capability of these coefficients in solving optimization problems. The search direction formula of the CG method in this study is defined as:

$$d_k = \begin{cases} -g_k, k = 0 \\ -g_k + \beta_k d_{k-1} - \theta_k y_{k-1}, k \geq 1 \end{cases} \tag{1.1}$$

where,

$$y_{k-1} = g_k - g_{k-1} \tag{1.2}$$

and,

$$\theta_k = \frac{g_k^T d_{k-1}}{d_{k-1}^T (d_{k-1} - g_k)} \tag{1.3}$$

The chosen CG coefficients are stated below.

Table 1.1

The List of CG coefficients

No	Author(s)	Coefficient Formula
1	Rivaie, Mustafa, Ismail & Leong	$\beta_k^{RMIL} = \frac{g_k^T (g_k - g_{k-1})}{\|d_{k-1}\|^2}$

2	Linda, Aini, Mustafa & Rivaie	$\beta_k^{LAMR} = \dfrac{g_k^T \left(\dfrac{\|d_{k-1}\|}{\|d_{k-1} - g_k\|} g_k - g_{k-1} \right)}{\dfrac{\|d_{k-1}\|}{\|d_{k-1} - g_k\|} \|d_{k-1}\|^2}$
3	Kamilu, Mustafa, Abdelrahman & Rivaie	$\beta_k^{KMAR} = \dfrac{g_k^T (g_k - g_{k-1})}{g_{k-1}^T (g_k + g_{k-1})}$
4	Sulaiman, Mustafa, Abdelrahman & Rivaie	$\beta_k^{SMAR} = \dfrac{g_k^T \left(g_k - \dfrac{\|g_k\|}{\|g_{k-1}\|} d_{k-1} \right)}{d_{k-1}^T d_{k-1}}$
5	Saleh, Fadhilah & Aidya	$\beta_k^{SFA} = -\dfrac{g_k^T \left(g_k - \dfrac{\|g_k\|}{\|g_{k-1}\|} g_{k-1} \right)}{d_{k-1}^T g_{k-1}}$

For the numerical test, the five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, are implemented using MATLAB R2022b subroutine programming. These TTCG methods are selected because previous studies have demonstrated their effectiveness in solving unconstrained optimization problems. MATLAB R2022b is used because it provides a suitable programming environment for implementing optimization algorithms and performing numerical computations efficiently.

The five TTCG methods are tested on fifteen standard unconstrained optimization test functions from the Constrained and Unconstrained Testing Environment (CUTE) library. The test functions are solved using different numbers of variables and different initial points. The initial points range from locations close to the minimizer to locations far from the minimizer, while the number of variables ranges from 2 to 10,000.

The numerical performance of the TTCG methods is evaluated by comparing the NOI and the CPU time required to reach the minimizer. The numerical results are then presented using performance profile graphs generated with SigmaPlot software. The results are analysed to compare the efficiency and robustness of the five TTCG methods. Lastly, the applicability of the best TTCG coefficient is demonstrated by

implementing it on an IMRT fluence map optimization problem. A clinical prostate dataset is used to determine the beamlet intensities that deliver the prescribed dose to the tumour while limiting the radiation dose to the rectum. However, the scope of this project is limited to examining the computational performance of the TTCG method and its coefficient variants, without considering clinical implementation or patient-specific treatment planning.

1.8 Project Benefits

This study demonstrates the potential of TTCG methods for solving large-scale optimization problems, particularly in IMRT. The findings provide a better understanding of the computational performance of different TTCG methods and their suitability for solving IMRT inverse planning problems.

The comparison of the five TTCG methods helps identify the method that provides the best computational performance in terms of the NOI and CPU time. The results also provide useful information for researchers in selecting an appropriate TTCG method for solving large-scale unconstrained optimization problems and IMRT fluence map optimization.

Furthermore, the implementation of the TTCG methods in IMRT demonstrates their practical application in radiotherapy treatment planning. An efficient TTCG method can reduce the computational time required to generate a treatment plan while maintaining good optimization performance. Therefore, this study contributes to the development of more efficient gradient-based optimization methods for IMRT and provides a useful reference for future research on large-scale optimization problems.

1.9 Definition of Terms and Concept

Several important terms and concepts are used throughout this study. The definitions are presented in this section to provide a clear reference for the discussions in the following chapters.

Table 1.2

Definition of Terms and Concept

Terms and Concepts	Definition
Optimization	The process of finding the minimum or maximum value of an objective function subject to constraints by adjusting decision variables (Ehrgott et al., 2008)
Conjugate Gradient (CG) Method	An efficient iterative optimization method that solves unconstrained optimization problems and large-scale systems while requiring only low memory (Tang et al., 2025).
Intensity-Modulated Radiation Therapy (IMRT)	Intensity-modulated radiation therapy (IMRT) is an advanced high-precision radiotherapy technique that uses computer-controlled linear accelerators to modulate radiation beam intensities, enabling highly conformal dose delivery to tumors while minimizing exposure to surrounding normal tissues (Moreno et al., 2025).
Rivaie-Mustafa-Ismail-Leong (RMIL)	RMIL coefficient is a CG parameter which alters the Polak-Ribière concept by using the last search direction in the denominator, and attempts to enhance descent behaviour and global convergence, in particular when using exact line search (Idalisa et al., 2023).
Linda-Aini-Mustafa-Rivaie (LAMR)	A modification method within the CG framework and is designed for solving unconstrained optimization problems and is proven to satisfy sufficient descent and global convergence properties (Zullpakkal et al., 2017).
Kamilu-Mustafa-Abdelrahman-Rivaie (KMAR)	A modified CG parameter, known as the KMAR coefficient, which is based on restart properties of classical CG methods but with a new denominator to ensure that descent and global convergence occur under exact line search (Kamfa et al., 2015).
Sulaiman-Mustafa-Abdelrahman-Rivaie (SMAR)	A modified β_k coefficient incorporating previous search direction information. Compared to classical methods such as Fletcher–Reeves and Polak-Ribiere-Polyak, SMAR demonstrates improved numerical stability and efficiency, especially for large-scale and nonconvex optimization problems (Mohammed et al., 2015).

Terms and Concepts	Definition
Saleh-Fadhilah-Aidya (SFA)	A three-term CG coefficient that extends the WYL-type formulation by scaling the gradient information to improve the search direction. Under exact line search, SFA satisfies the sufficient descent condition and achieves global convergence, showing competitive efficiency and robustness across standard test problems (Saleh Nazzal et al., 2023).
Step Size, α_k	A scalar parameter that determines the length of the update along the search direction in an iterative optimization algorithm, selected using a line search strategy to ensure stable convergence (Nocedal & Wright, 2006).
Search direction, d_k	A direction vector along which the optimization algorithm updates the current solution at iteration k, determined using gradient information and CG coefficients (Stanimirović et al., 2020).
Organ at Risk (OARs)	Critical organs are normal anatomical structures surrounding the target volume for which radiation dose must be minimized to prevent serious complications (Maass et al., 2022).
Inverse Planning	A computational optimization process in IMRT in which optimization algorithms are used to determine beamlet intensities based on prescribed dose objectives and clinical constraints (Bortfeld, 1999).

1.10 Research Outline

This research is structured systematically to give a clear outline of the research. There are six chapters, which include the introduction, the literature review, the methodology, implementation, results and discussion, and lastly the conclusion and recommendation. Chapter One introduces the research by providing the research background, problem statement, research objectives and the significance of the research. It also defines the scope and benefits of the research, the definition of the term and concept and lastly presents an overview of the research outline. Chapter Two presents a comprehensive literature review that examines previous research and existing knowledge on the research topic.

Next, Chapter Three describes the research methodology, which includes the chosen CG coefficient. The explanation of the research step is provided. Chapter Four focuses on the implementation phase, where the proposed methods and application are developed and applied. All fifteen test functions are presented in detail in this chapter. Chapter Five discusses the results obtained from the implementation and provides a detailed analysis of the findings. Finally, Chapter Six concludes the report by summarizing the main conclusions of this research and providing relevant recommendations for future research or practical applications.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter presents the literature review. The purpose of the literature review is to provide an overview of existing studies by previous researchers, theories and methodologies relevant to the CG method. This chapter gives an overview of the current state of knowledge, identifies the research gaps, and sets the context for this research. The sources used in the literature review are journal articles and websites.

2.2 Literature Review

This section presents an overview of the fundamental concepts related to optimization methods, with a focus on the CG method and its classification. Finally, the IMRT inverse problem is discussed.

2.3 Optimization

Optimization is a mathematical process for finding the best solution among a set of possible choices. In mathematical modelling, an optimization problem is formulated as an objective function that is minimized or maximized while achieving the required objectives without violating the specified constraints (Deng et al., 2025). In general, the problem is to find the minimum or maximum value of an objective function subject to given conditions, and it is written as:

$$\min f(x), x \in \mathbb{R}^n \tag{2.1}$$

Where $f : \mathbb{R}n \to \mathbb{R}$ is a continuously differentiable objective function and x is an element of the real numbers (Montgomery, 2025). Several optimization methods have been proposed in the literature for solving unconstrained optimization problems, including the Steepest Descent (SD) method, Newton's method, Quasi-Newton (QN) methods, and the Conjugate Gradient (CG) method.

The SD method is simple to implement but generally converges slowly, especially for ill-conditioned and large-scale optimization problems. Newton's method and QN methods usually achieve faster convergence. However, both methods require the computation and storage of the Hessian matrix or its approximation, resulting in higher computational cost and memory requirements (Nocedal & Wright, 2006). Therefore, these methods are often unsuitable for large-scale optimization problems such as IMRT inverse planning.

In contrast, the CG method requires only first-order derivative information and has low memory requirements. These advantages make the CG method computationally efficient and well suited for solving large-scale optimization problems (Hager et al., 2006; Nocedal & Wright, 2006).

Table 2.1
List of Optimization Methods

No.	Method	Search Direction
1	Steepest Descent	$d_k = -g_k$
2	Newton's	$d_k = -H_k^{-1} g_k$
3	Quasi-Newton	$d_k = -H_k g_k$
4	Conjugate Gradient	$d_k = -g_k + \beta_k d_{k-1}$

The search direction formulas presented in Table 2.1 show how each optimization method determines the direction of movement at every iteration. The SD method uses only the negative gradient, Newton's method uses the inverse Hessian matrix, the QN method uses an approximation of the inverse Hessian matrix, and the CG method combines the negative gradient with the previous search direction through the CG coefficient. These different formulations influence the convergence behaviour, computational cost, and memory requirements of each method.

Among these methods, the CG method is particularly suitable for large-scale optimization problems because it requires only first-order derivative information and does not require the computation or storage of the Hessian matrix. As a result, the CG method has low memory requirements while maintaining good computational

efficiency (Hager et al., 2006; Nocedal & Wright, 2006). These advantages make the CG method a suitable choice for solving large-scale optimization problems such as IMRT inverse planning.

2.4 Conjugate Gradient Method

The CG method was developed to improve the computational efficiency and convergence of the SD method for solving unconstrained optimization problems, particularly large-scale problems. The SD method updates the solution by moving in the direction of the negative gradient, which represents the direction of the steepest decrease of the objective function. Although the SD method is simple to implement, the search direction is determined solely by the current gradient. Consequently, the search direction changes at every iteration, often producing a zigzag search path, especially for ill-conditioned problems. This behaviour results in slow convergence and increased computational cost (Kiran, 2022). To overcome this limitation, the CG method combines the current gradient with the previous search direction to generate conjugate directions. This allows the algorithm to maintain a more consistent search path, reduce zigzagging, and achieve faster convergence than the SD method, especially for large-scale unconstrained optimization problems (Fletcher & Reeves, 1964; Hestenes & Stiefel, 1952). Hence, CG iterative method is expressed as:

$$x_{k+1} = x_k + \alpha_k d_k, k = 0,1,2,3 \tag{2.2}$$

Where $\alpha_k > 0$ is a step size determined by a line search strategy, d_k is the search direction and x_k represent the current iteration point. Other researchers have demonstrated that the choice of search direction and step size selection procedure have a strong influence on the computational performance of the CG method.

Consequently, various line search methods, such as exact and inexact line searches, have been explored with a view to providing adequate descent and numerical stability (Nocedal & Wright, 2006). One of the most widely used line search techniques is the exact line search, which determines the step size by minimizing the objective function exactly along the search direction. This allows the algorithm to achieve the maximum reduction in the objective function at each iteration. Although the exact line

search generally provides good convergence and stable search directions, it requires solving a one-dimensional optimization problem at every iteration. As a result, the exact line search is computationally expensive, particularly for large-scale optimization problems.

2.4.1 Conjugate Gradient Coefficients

One of the important factors of the CG method is the choice of CG coefficients, β_k, that determines how the conjugate search directions are built. Over the years, numerous CG coefficients have been proposed to improve computational performance and numerical robustness.

Table 2.2
Conjugate Gradient Coefficients

No.	Author(s)	Year	Coefficient Formula
1	Hestenes & Stiefel	1952	$\beta_k^{HS} = \frac{g_k^T (g_k - g_{k-1})}{d_{k-1}^T (g_k - g_{k-1})}$
2	Fletcher & Reeves	1964	$\beta_k^{FR} = \frac{g_k^T g_k}{g_{k-1}^T g_{k-1}}$
3	Polak, Ribière & Polyak	1969	$\beta_k^{PRP} = \frac{g_k^T (g_k - g_{k-1})}{g_{k-1}^T g_{k-1}}$
4	Liu & Storey	1991	$\beta_k^{LS} = \frac{-g_k^T g_{k-1}}{d_{k-1}^T (g_k - g_{k-1})}$
5	Dai & Yuan	1999	$\beta_k^{DY} = \frac{\|g_k\|^2}{d_{k-1}^T (g_k - g_{k-1})}$
6	Rivaie, Mustafa, Ismail and Leong	2012	$\beta_k^{RMIL} = \frac{g_k^T (g_k - g_{k-1})}{\|d_{k-1}\|^2}$

Even though several CG coefficients have been demonstrated to work well in numerical experiments, some classical CG algorithms can be inefficient or be deficient in descent

properties in some cases, especially when used to solve large scale optimization problems (Hager et al., 2006).

2.4.2 Three-Term Conjugate Gradient

The classical two-term CG method generates a search direction of the form $d_k = g_k + \beta_k d_{k-1}$, where the scalar β_k is the conjugate coefficient that ensures the new direction is conjugate to all previous directions. Although this formulation is simple and computationally inexpensive, it may fail to maintain the sufficient descent property for nonconvex functions and can exhibit slow convergence or reduced efficiency in practice, particularly when the iterates encounter nearly flat regions or when an inexact line search is used (Hager et al., 2006; Powell, 1984).

The TTCG method is introduced to overcome these limitations by adding an extra term to the search direction formula. The general TTCG search direction is written as:

$$d_k = -g_k + \beta_k d_{k-1} - \theta_k y_{k-1}, k \geq 1; \tag{2.3}$$

where y_{k-1} is the gradient difference vector as stated in Equation (1.2), β_k is the conjugate coefficient from any classical CG method, and θ_k is an extra scalar parameter defined to satisfy the sufficient descent condition:

$$g_k^T d_k \leq -c\|g_k\|^2 \tag{2.4}$$

for some constant $c > 0$. In the formulation used in this study, θ_k is defined as Equation (1.3).

This formulation guarantees the sufficient descent property regardless of the line search. This is an advantage over classical two-term methods, which need specific line search conditions to keep the descent property (Baluch et al., 2017; Narushima et al., 2011). The TTCG framework is first conceptualised by Beale in 1972 as a restart strategy (Aminifard & Babaie-Kafaki, 2020). Zhang et al. (2007) later extended the three-term Hestenes-Stiefel method with a strong Wolfe line search and showed that the method reduces to the standard CG method for strictly convex quadratic functions.

Additionally, TTCG methods have several advantages. The sufficient descent property holds without any condition on the line search. The global convergence is stronger for general nonconvex functions. The numerical behaviour is also more stable because extra gradient information is carried in the third term (Liu et al., 2018; Sulaiman et al., 2022). These properties make TTCG methods suitable for large-scale and possibly ill-conditioned problems such as IMRT inverse planning.

In this study, the TTCG framework is implemented using five CG coefficient formulations, namely RMIL (Idalisa et al., 2022), LAMR (Zullpakkal et al., 2024), KMAR (Kamilu et al., 2023), SMAR (Mohammed et al., 2015), and SFA (Saleh Nazzal et al., 2023) resulting in the TTCG methods TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, respectively. Each TTCG method uses a different CG coefficient to construct the search direction, which may lead to differences in convergence and computational performance. Therefore, a comparative study of the five TTCG methods is carried out using a common exact line search to evaluate their performance under the same optimization framework.

2.5 Exact Line Search

In iterative optimization, the step size, α_k is determined using a procedure known as a line search. The exact line search computes the step size that minimizes the objective function exactly along the current search direction, d_k, that is:

$$\alpha_k = \arg\min_{\alpha>0} f(x_k + \alpha d_k) \tag{2.5}$$

This is equivalent to finding the point where the derivative of the objective function along the search direction is zero. Therefore, the exact line search satisfies the first-order optimality condition

$$\nabla f(x_k + \alpha_k d_k)^T d_k = 0 \tag{2.6}$$

For quadratic objective functions, the exact line search ensures that successive search directions generated by the CG method are conjugate, allowing the algorithm to converge in at most n iterations for an *n*-dimensional problem (Hestenes & Stiefel, 1952; Nocedal & Wright, 2006).

Although the exact line search generally provides good convergence and stable search directions, it requires solving a one-dimensional optimization problem at every iteration. As a result, the exact line search is computationally expensive, particularly for large-scale optimization problems (Nocedal & Wright, 2006).

Despite this limitation, the exact line search is widely used in numerical experiments because it provides a fair and consistent basis for comparing different CG methods. By computing the optimal step size at every iteration, the performance of different CG methods can be evaluated without the influence of step-length approximation errors. Therefore, the exact line search is adopted in this study to compare the performance of the selected TTCG methods under the same optimization framework.

2.6 Intensity-Modulated Radiation Therapy (IMRT)

IMRT treatment planning is typically formulated as an inverse optimization problem, where the aim is to determine beamlet intensities that adequately cover the target while keeping the dose to the OARs within acceptable limits. The optimization problem involves thousands of variables and constraints, making the efficiency of the optimization algorithm essential for clinical applications (Cho, 2018). In addition, IMRT optimization requires repeated evaluation of the objective function and its gradient throughout the treatment planning process.

Since IMRT optimization is a computationally challenging problem, the selection and application of effective algorithms is essential to make such planning clinically feasible. In practice, an algorithm may need a large NOI before it converges, and even small inefficiencies in each iteration can greatly increase the total planning time for a patient. For this reason, the optimization method is a key factor in producing IMRT plans efficiently and reliably.

Among the available approaches, gradient-based methods are widely used in IMRT planning because they can handle large-scale problems without needing to compute or store second-order derivative information. Within this group, the CG method has shown stronger computational performance than the SD method (Fletcher & Reeves, 1964; Hestenes & Stiefel, 1952). The CG method has also been applied directly to IMRT fluence map optimization, where Ólafsson et al. (2005) used the CG algorithm to compute the Newton component of the search direction in EUD-based

planning and found it to be the most effective of the variants they tested. The reason the CG method suits IMRT lies in the mathematical structure of the problem. Alber et al. (2002) showed that the IMRT objective function is highly degenerate, remaining nearly flat in most directions and curving sharply only within a small subspace. Since the number of CG iterations depends on the number of distinct eigenvalues of the Hessian matrix rather than the number of variables, the method can converge in relatively few iterations even when the number of beamlets is large.

The CG method has also been used inside hybrid frameworks for IMRT, not only as a standalone solver. Cao et al. (2022) used the CG method as a local search operator inside a multi-objective evolutionary algorithm for fluence map optimization. The CG step speeds up the convergence of the population while a diverse set of Pareto-optimal treatment plans is produced in a single run. In five clinical cancer cases, the CG-assisted approach performed better than the traditional weighted-sum CG method and several other multi-objective evolutionary algorithms, further confirming the continued relevance and efficiency of the CG method in IMRT planning.

However, the performance of the CG method depends strongly on the CG coefficient used to build the search direction. Different coefficients give different numbers of iterations and computational times. Even though computational efficiency is important in IMRT planning, few studies have compared how different CG coefficients affect the NOI and CPU time.

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2.7 Conclusion

In conclusion, this chapter reviews the relevant literature related to the research topic and identifies the gap concerning the limited comparison of TTCG coefficient variants under exact line search in IMRT optimization. The knowledge gained provides a strong foundation for the current research and justifies the need for further investigation, as described in the subsequent chapters.

CHAPTER 3
RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains the methodology stages to achieve the research objectives and to solve the problem statements. The methodology begins with project definition, followed by literature review, numerical test and numerical analysis. Next, the application of the TTCG coefficient is discussed. Lastly, the methodology concludes with a discussion and conclusion addressing the overall research aims.

3.2 Research Step

This study is carried out through seven main steps to ensure that the research objectives are achieved systematically. Figure 3.1 shows the six research steps of this study in finding solutions.

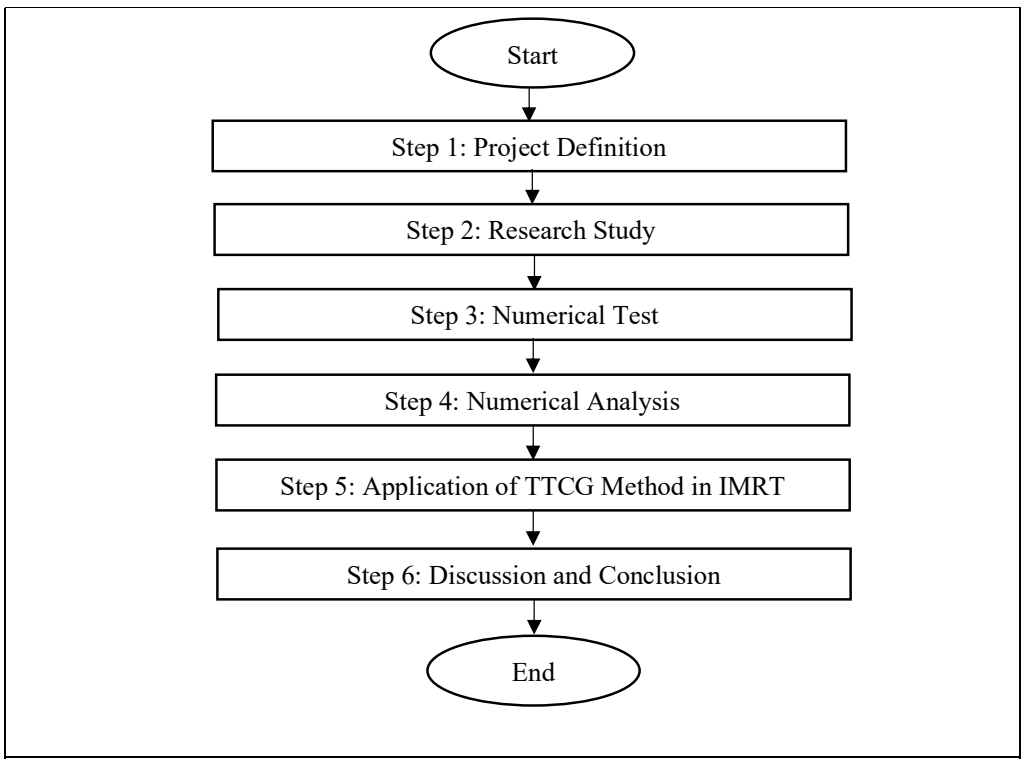


Figure 3.1 Research Methodology Flowchart

Step 1: Project Definition

This study focuses on the implementation and comparison of five TTCG methods for solving large-scale optimization problems and their application to IMRT. The main objective is to identify the most suitable TTCG method based on computational performance. The comparison is carried out using the NOI, CPU time, and robustness.

Step 2: Research Study

A comprehensive literature review is conducted on unconstrained optimization, the CG method, TTCG methods, exact line search, and IMRT fluence map optimization. Previous studies on the five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, are also reviewed to understand their mathematical formulation and computational properties.

Step 3: Numerical Test

The numerical experiments are carried out using MATLAB R2022b. The five TTCG methods are implemented and tested on fifteen standard unconstrained optimization test functions from the CUTE library. The test problems are solved using different initial points and numbers of variables ranging from 2 to 10,000.

Step 4: Numerical Analysis

The numerical results obtained from the standard test functions are analysed by comparing the performance of the five TTCG methods in terms of the NOI and CPU time. Performance profile graphs are also generated to evaluate the efficiency and robustness of each TTCG method. The analysis is then used to identify the most suitable TTCG method for solving standard unconstrained optimization problems.

Step 5: Application of TTCG Method in IMRT

All five TTCG methods are implemented in an IMRT fluence map optimization problem using a clinical prostate dataset. The performance of the TTCG methods is compared in terms of the number of iterations (NOI), CPU time, final objective value, and the percentage of rectum volume receiving more than 30 Gy. The comparison is used to evaluate the suitability of each TTCG method for IMRT inverse planning.

Step 6: Discussion and Conclusion

The findings from both the numerical experiments and the IMRT application are discussed and interpreted. The performance of the TTCG methods is compared based on the selected performance measures, and the overall results are used to identify the most suitable TTCG method for IMRT. Finally, conclusions are drawn, and recommendations for future research are presented.

3.3 MATLAB Algorithm

The MATLAB R2022b algorithm is structured into six steps as described below. The algorithm is applied to all five TTCG coefficients, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR and TTSFA.

Step 1: Initialize the initial point, x_k and set $k = 0$.

Step 2: Compute the TTCG coefficients, β_k which are TTRMIL, TTLAMR, TTKMAR TTSMAR and TTSFA as listed in Table 1.1.

Step 3: Compute the three-term search direction, d_k based on Equation (1.1):

$$d_k = \begin{cases} -g_k, k = 0 \\ -g_k + \beta_k d_{k-1} - \theta_k y_{k-1}, k \geq 1 \end{cases}$$

where y_{k-1} and θ_k are computed using Equation (1.2) and (1.3), respectively.
Stop if $\|g_k\| = 0$.

Step 4: Based on Equation (2.5), compute the step length, α_k using exact line search.

Step 5: Update the new point based on the iterative using Equation (2.2).

Step 6: Test the convergence and check the stopping criterion. Stop if $\|g_k\| \leq 10^{-6}$. Otherwise, return to Step 2 with $k = k + 1$.

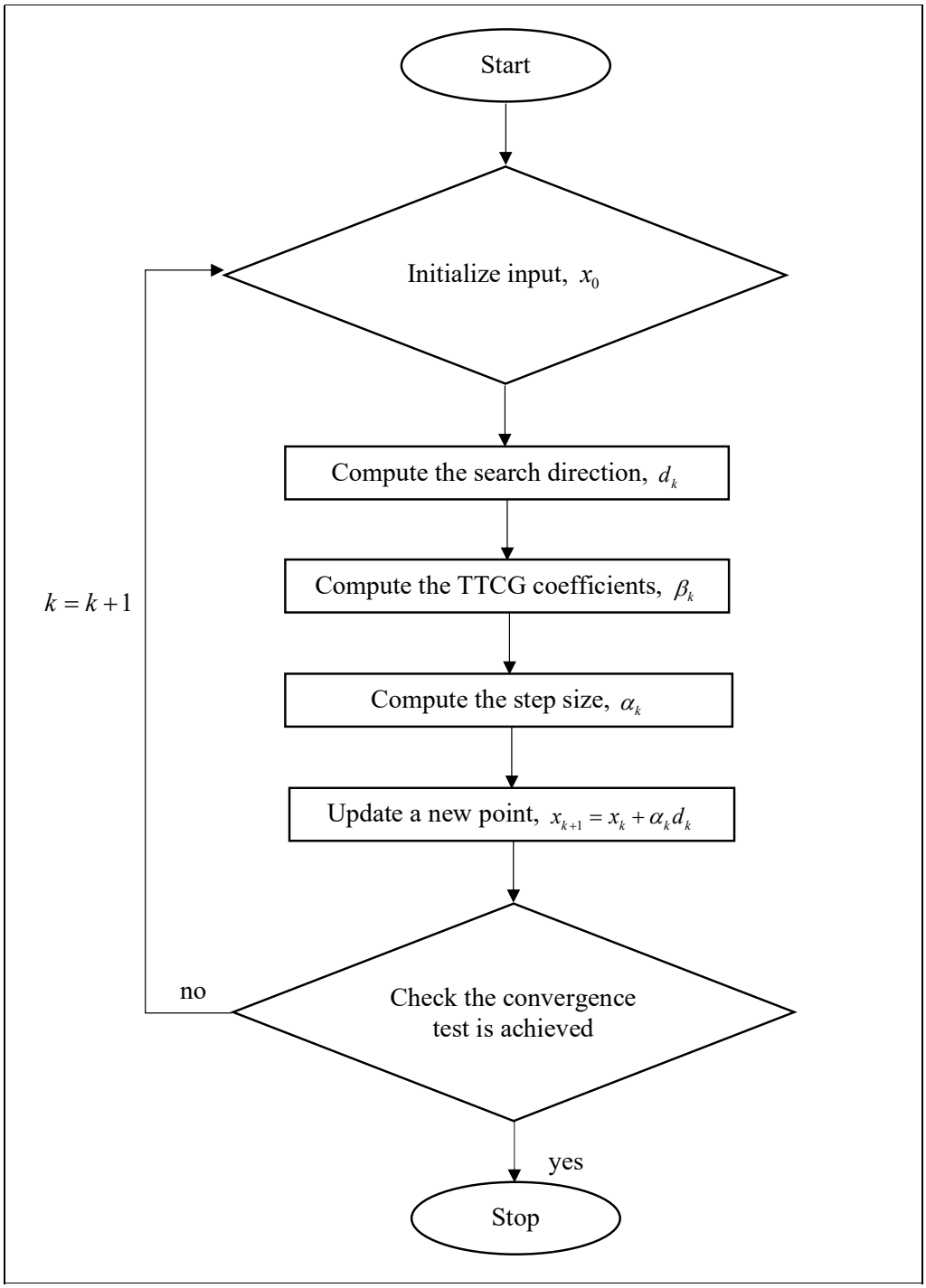


Figure 3.2 MATLAB Methodology Flowchart

3.4 Theoretical Calculations

A theoretical calculation is a mathematical computation that uses established scientific theories and principles to determine a value without direct experimental measurement. This section presents a manual step-by-step calculation of the TTCG method using the TTLAMR coefficient, applied to the Extended Himmelblau test function. The purpose is to demonstrate how the algorithm operates and progresses from an initial point towards the minimum.

The function selected for this demonstration is the Extended Himmelblau function, which for $n = 2$ is expressed as Equation (4.12):

$$f(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

The initial point selected is $x_0 = (2,2)^T$, which is the first initial point listed in Table 4.1 for the Extended Himmelblau function. The global minimum is $f(x^*) = 0$, achieved at several points. The convergence criterion is $\|g_k\| \leq 10^{-6}$.

The gradient of the function is obtained by differentiating f with respect to each variable:

$$\frac{\partial f}{\partial x_1} = 4x_1(x_1^2 + x_2 - 11) + 2(x_1 + x_2^2 - 7)$$

$$\frac{\partial f}{\partial x_2} = 2(x_1^2 + x_2 - 11) + 4x_2(x_1 + x_2^2 - 7)$$

The TTLAMR coefficient formula used in this research is:

$$\beta_k^{TTLAMR} = \frac{g_k^T \left(\frac{\|d_{k-1}\|}{\|d_{k-1} - g_k\|} g_k - g_{k-1} \right)}{\frac{\|d_{k-1}\|}{\|d_{k-1} - g_k\|} \|d_{k-1}\|^2}$$

Based on Equation (1.1), the three-term search direction is defined as:

$$d_k = -g_k + \beta_k d_{k-1} - \theta_k y_{k-1} \; if \; k \geq 1;$$

which $y_{k-1} = g_k - g_{k-1}$ and $\theta_k = \dfrac{g_k^T d_{k-1}}{d_{k-1}^T (d_{k-1} - g_k)}$ for $k \geq 1$

The step length, α_k is determined using exact line search in Equation (2.5):

$$\alpha_k = \arg\min_{\alpha > 0} f\left(x_k + a d_k\right)$$

Below are the complete steps of the manual calculation for the first two iterations.

Step 1: Initialization

Initialize the starting point and set $k = 0$:

$$x_0 = (2,2)^T, k = 0$$

Compute the function value at x_0:

$$f(x_0) = (2^2 + 2 - 11)^2 + (2 + 2^2 - 7)^2 = 26$$

Compute the gradient $g_0 = \nabla f(x_0)$ by substituting $x_1 = 2$ and $x_2 = 2$:

$$\frac{\partial f}{\partial x_1} = 4(2)(4 + 2 - 11) + 2(2 + 4 - 7) = -42$$

$$\frac{\partial f}{\partial x_2} = 2(4 + 2 - 11) + 4(2)(2 + 4 - 7) = -18$$

$$g_0 = [-42, -18]^T$$

Compute the norm of the gradient:

$$\|g_0\| = \sqrt{(-42)^2 + (-18)^2} \approx 45.6946$$

Since $\|g_0\| \approx 45.6946 \geq 10^{-6}$, the stopping criterion is not satisfied. The algorithm proceeds.

Step 2: TTLAMR Coefficient Computation

For the first iteration $(k = 0)$, the TTLAMR coefficient is not yet computed since there is no previous iteration data. Therefore:

$$\beta_0 = 0$$

The TTLAMR coefficient is computed from the second iteration onwards, once previous gradient and direction information becomes available $(k = 1)$.

Step 3: Search Direction Computation (k = 0)

For $k = 0$, since $\beta_0 = 0$, the three-term search direction reduces to the steepest descent direction:

$$d_0 = -g_0 = -[-42, -18]^T = [42, 18]^T$$

Verify that this is a descent direction by checking $g_0^T d_0 < 0$:

$$g_0^T d_0 = (-42)(42) + (-18)(18) = -2088$$
$$\therefore g_0^T d_0 < 0$$

The descent condition is satisfied.

Step 4: Step Length Computation (k = 0)

The exact line search minimizes $f(\alpha) = f(x_0 + \alpha d_0)$ over $\alpha > 0$. Substituting:

$$x_0 + \alpha d_0 = (2 + 42\alpha, 2 + 18\alpha)^T$$

Substituting into the Extended Himmelblau function:

$$f(\alpha) = ((2 + 42\alpha)^2 + (2 + 18\alpha) - 11)^2 + ((2 + 42\alpha) + (2 + 18\alpha)^2 - 7)^2$$

Expanding and simplifying:

$$f(\alpha) = 3216672\alpha^4 + 730080\alpha^3 + 29304\alpha^4 - 2088\alpha + 26$$

Set $\frac{df}{d\alpha} = 0$ to find the minimizer:

$$\frac{df}{d\alpha} = 12866688\alpha^3 + 2190240\alpha^2 + 58608\alpha - 2088 = 0$$

Solving this cubic equation numerically gives one real positive root. The minimum value of $f(\alpha)$ occurs at:

$$\alpha_0 \approx 0.019607$$

Step 5: Update the New Point (k = 0)

Using the iterative update formula $x_{k+1} = x_k \alpha_k d_k$:

$$x_1 = x_0 + \alpha_0 d_0 = (2,2)^T + 0.019607 \times (42,18)^T = (2.8235, 2.3529)^T$$

Compute the function value and gradient at the new point x_1 :

$$f(x_1) = (2.8235^2 + 2.3529^2 - 11)^2 + (2.8235 + 2.3529^2 - 7)^2 \approx 2.3045$$

Computing the gradient $g_1 = \nabla f(x_1)$:

$$g_1 = \left[-4.9047, 11.4469\right]^T$$

$$\|g_1\| = \sqrt{(-4.9047)^2 + (11.4469)^2} \approx 12.4534$$

Since $\|g_1\| \approx 12.4534 > 10^{-6}$, the algorithm continues to $k = 1$.

Step 6: TTLAMR Coefficient Computation (k = 1)

Now that previous iteration data is available, the TTLAMR coefficient is computed using the formula:

$$\beta_1^{TTLAMR} = \frac{g_1^T \left(\frac{\|d_0\|}{\|d_0 - g_1\|} g_1 - g_0 \right)}{\frac{\|d_0\|}{\|d_0 - g_1\|} \|d_0\|^2}$$

First, compute the required norms:

$$\|d_0\| = \sqrt{(42)^2 + (18)^2} \approx 45.6946$$

$$d_0 - g_1 = \left[42 - (-4.9047), 18 - 11.4469\right]^T = \left[46.9047, 6.5531\right]^T$$

$$\|d_0 - g_1\| = \sqrt{(46.9047)^2 + (6.5531)^2} \approx 47.3603$$

Complete the ratio:

$$\frac{\|d_0\|}{\|d_0 - g_1\|} = \frac{45.6946}{47.3603} \approx 0.9648$$

Compute the numerator vector inside the bracket:

$$\frac{\|d_0\|}{\|d_0 - g_1\|}(g_1 - g_0) = 0.9648 \times [-4.9047, 11.4469]^T = [37.2675, 29.0444]^T$$

Compute the numerator (dot product with g_1):

$$g_1^T \times [37.2675, 29.0444]^T = (-4.9047)(37.2675) + (11.4469)(29.0444) = 149.669$$

Compute the denominator:

$$\frac{\|d_0\|}{\|d_0 - g_1\|}\|d_0\|^2 = 0.9648 \times (45.6946)^2 \approx 2014.55$$

Therefore:

$$\beta_1^{TTLAMR} = \frac{149.669}{2014.55} \approx 0.0743$$

Step 7: Three-Term Search Direction (k = 1)

Compute the gradient difference vector:

$$y_0 = g_1 - g_0 = [-4.9047 - (-42), 11.4469 - (-18)]^T = [37.0953, 29.4469]^T$$

Compute the θ_1 parameter:

$$\theta_1 = \frac{g_1^T d_0}{d_0^T (d_0 - g_1)}$$

$$g_1^T d_0 = (4.9047)(42) + (11.4469)(18) = 0.047$$

$$d_0^T (d_0 - g_1) = (42)(46.9047) + (18)(6.5531) = 2087.953$$

$$\theta_1 = \frac{0.047}{2087.953} \approx 0.000022$$

Compute the three-term search direction:

$$d_1 = -g_1 + \beta_1 d_0 - \theta_1 y_0$$

$$d_1 = -\left[-4.9047, 11.4469\right]^T + 0.0743 \times \left[42, 18\right]^T - 0.000022 \times \left[37.0953, 29.4469\right]^T$$

$$\approx \left[8.0245, -10.1101\right]^T$$

Verify the descent condition $g_1^T d_1 < 0$:

$$g_1^T d_1 = (-4.9047)(8.0245) + (11.4469)(-10.1101) = -155.088 < \mathbf{0}$$

The sufficient descent condition is satisfied.

Step 8: Step Length and Update (k = 1)

Applying the exact line search to minimize $f(x_1 + \alpha d_1)$ over $\alpha > 0$:

$$\alpha_1 = 0.027000$$

$$x_2 = x_1 + \alpha_1 d_1$$

$$= (2.8235, 2.3529)^T + 0.027000 \times (8.0245, -10.1101)^T$$

$$= (3.0402, 2.0799)^T$$

$$f(x_2) \approx 0.2381$$

$$\|g_2\| \approx 5.9396$$

Since $\|g_2\| \approx 5.9396 > 10^{-6}$, the algorithm continues with $k = 2$

Step 9: Check Convergence and Stopping Criterion

Stop if $\|g_k\| \le 10^{-6}$. Otherwise, return to Step 2 with $k = k + 1$ until the stopping criterion is reached.

Table 3.1
Summary of TTLAMR iterations for the Extended Himmelblau function $(x_0 = (2,2), n = 2)$

Iteration, k	$\mathbf{x_k}$	$\mathbf{f}(\mathbf{x_k})$	$\|\mathbf{g_k}\|$	$\boldsymbol{\alpha_k}$
0	(2,2)	26.0000	45.6946	0.019607
1	(2.8235, 2.3529)	2.3045	12.4534	0.027000
2	(3.0402, 2.0799)	0.2381	5.9396	-

The table confirms a consistent reduction in both $f(x_k)$ and $\|g_k\|$ across iterations. The algorithm will continue iterating until $\|g_k\| \le 10^{-6}$, at which point it converges to the global minimum of the Extended Himmelblau function at $f(x^*) = 0$.

3.5 Conclusion

In conclusion, this chapter describes the research methodology used to achieve the objectives of this study. The research is carried out in six main steps, beginning with the project definition and research study, followed by the numerical test, numerical analysis, application of the TTCG methods in IMRT, and finally the discussion and conclusion. The MATLAB algorithm of the TTCG method under the exact line search is also presented step by step, starting from the initialization of the initial point until the stopping criterion is satisfied.

In addition, a manual calculation of the TTCG method using the TTLAMR method on the Extended Himmelblau test function is presented to illustrate the implementation of the algorithm at each iteration. The calculation shows that the sufficient descent condition is satisfied and that the objective function value decreases throughout the optimization process until convergence is achieved. In the next chapter, the proposed methodology is implemented on fifteen standard unconstrained optimization test functions. The performance of the five TTCG methods is then evaluated in terms of the NOI, CPU time, and performance profiles before being applied to an IMRT fluence map optimization problem.

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CHAPTER 4
IMPLEMENTATION

4.1 Introduction

This chapter presents the implementation of the five TTCG methods, which are TTRMIL, TTLAMR, TTKMAR, TTSMAR and TTSFA. It first presents the fifteen standard unconstrained test functions together with their mathematical formulas, initial points, and numbers of variables. It then describes the performance profile as the evaluation tool for comparing the efficiency and robustness of the coefficients, and formulates the IMRT fluence map optimization problem, including the mathematical model and the clinical prostate case used in this study.

4.2 Numerical Test

The five TTCG methods are tested by solving fifteen standard test functions taken from the CUTE library. All experiments are run in MATLAB R2022b using exact line search. Each method stops when the gradient norm satisfies the condition $\|g_k\| \leq 10^{-6}$. The methods are then compared using performance profiles based on NOI and CPU time.

4.2.1 Test Functions

A total of fifteen test functions is selected to test the performance of all five TTCG methods. The functions are tested with different initial points and different numbers of variables such as 2, 4, 10, 100, 500, 1000, and 10,000 in order to check how well each method works under various conditions. Table 4.1 shows the list of test functions, initial points, and the number of variables used.

Table 4.1
List of Test Functions with Initial Points and Number of Variables

Test Functions	Initial Points	Number of Variables
Extended Maratos	(2,2), (5,5), (8,8), (11,11)	2, 4, 10
Power	(2,2), (5,5), (7,7), (11,11)	2, 4, 10
Perturbed Quadratic	(3,3), (7,7), (9,9), (13,13)	2, 4, 10
Raydan 1	(3,3), (8,8), (13,13), (18,18)	2, 4, 10
FLETCHCR	(5,5), (10,10), (15,15), (20,20)	2, 4, 10
Extended Penalty	(3,3), (7,7), (10,10), (23,23)	2, 4, 10, 100
Hager	(4,4), (8,8), (18,18), (21,21)	2, 4, 10, 100
Ext. White & Holst	(2,2), (5,5), (9,9), (11,11)	2, 4, 10, 100, 500, 1000
Ext. Tridiagonal 1	(2,2), (25,25), (35,35), (45,45)	2, 4, 10, 100, 500, 1000
Extended Rosenbrock	(2,2), (5,5), (7,7), (9,9)	2, 4, 10, 100, 500, 1000, 10,000
Shallow	(2,2), (7,7), (14,14), (20,20)	2, 4, 10, 100, 500, 1000, 10,000
Ext. Himmelblau	(2,2), (6,6), (14,14), (24,24)	2, 4, 10, 100, 500, 1000, 10,000
Sphere	(2,2), (8,8), (16,16), (24,24)	2, 4, 10, 100, 500, 1000, 10,000
Ext. DENSCHNB	(3,3), (8,8), (15,15), (21,21)	2, 4, 10, 100, 500, 1000, 10,000
Diagonal 4	(5,5), (10,10), (20,20), (30,30)	2, 4, 10, 100, 500, 1000, 10,000

All 15 test functions are implemented in MATLAB R2022b. The mathematical formula for each function is listed below.

1. Extended Maratos

$$f(x) = \sum_{i=1}^{\frac{n}{2}} x_{2i-1} + 100(x_{2i-1}^2 + x_{2i}^2 - 1)^2 \tag{4.1}$$

2. Power

$$f(x) = \sum_{i=1}^{n} (ix_i^2)^2 \tag{4.2}$$

3. Perturbed Quadratic

$$f(x) = \sum_{i=1}^{n} ix_i^2 + \frac{1}{100}\left(\sum_{i=1}^{n} x_i\right)^2 \tag{4.3}$$

4. Raydan 1

$$f(x) = \sum_{i=1}^{n} \frac{i}{10}(e^{x_i} - x_i) \tag{4.4}$$

5. FLETCHCR

$$f(x) = \sum_{i=1}^{n-1} 100\left(x_{i+1} - x_i + 1 - x_i^2\right)^2 \tag{4.5}$$

6. Extended Penalty

$$f(x) = \sum_{i=1}^{n-1} (x_i - 1)^2 + \sum_{j=1}^{n} \left(x_j^2 - 0.25\right)^2 \tag{4.6}$$

7. Hager

$$f(x) = \sum_{i=1}^{n} \left(e^{x_i} - \sqrt{i} x_i \right) \tag{4.7}$$

8. Extended White & Holst

$$f(x) = \sum_{i=1}^{\frac{n}{2}} \left[100 \left(x_{2i} + x_{2i-1}^3 \right)^2 + \left(1 - x_{2i-1} \right)^2 \right] \tag{4.8}$$

9. Extended Tridiagonal 1

$$f(x) = \sum_{i=1}^{\frac{n}{2}} \left(x_{2i-1} + x_{2i} - 3 \right)^2 + \left(x_{2i-1} - x_{2i} + 1 \right)^4 \tag{4.9}$$

10. Extended Rosenbrock

$$f(x) = \sum_{i=1}^{\frac{n}{2}} \left[100 \left(x_{2i} + x_{2i-1}^2 \right)^2 + \left(1 - x_{2i-1} \right)^2 \right] \tag{4.10}$$

11. Shallow

$$f(x) = \sum_{i=1}^{\frac{n}{2}} \left[\left(x_{2i}^2 - x_{2i} \right)^2 + \left(1 - x_{2i-1} \right)^2 \right] \tag{4.11}$$

12. Extended Himmelblau

$$f(x) = \sum_{i=1}^{\frac{n}{2}} \left[\left(x_{2i-1}^2 + x_{2i} - 11 \right)^2 + \left(x_{2i-1} + x_{2i}^2 - 7 \right)^2 \right] \tag{4.12}$$

13. Sphere

$$f(x) = \sum_{i=1}^{n} x_i^2 \tag{4.13}$$

14. Extended DENSCHNB

$$f(x)=\sum_{i=1}^{\frac{n}{2}}\left[\left(x_{2i-1}-2\right)^2+\left(x_{2i-1}-2\right)^2 x_{2i}+\left(x_{2i}+1\right)^2\right] \tag{4.14}$$

15. Diagonal 4

$$f(x)=\sum_{i=1}^{\frac{n}{2}}\frac{1}{2}\left(x_{2i-1}^2+100x_{2i}^2\right) \tag{4.15}$$

4.2.2 Performance Profile

The performance profile method by Dolan and Moré (2001) is used to compare and evaluate the results. This method shows, for each coefficient, what percentage of test problems it can solve within a certain cost compared to the best method. The cost here refers to either NOI or CPU time. The main idea is to compare all five TTCG methods across all test problems. For each problem, the best (lowest) cost among all methods is found first. This is written as:

$$\min\left\{t_{p,s}:s\in S\right\} \tag{4.16}$$

where $t_{p,s}$ denotes the cost (NOI and CPU time) of solver s on problem p. Then, the performance ratio for each method is calculated as:

$$r_{p,s}=\frac{t_{p,s}}{\min\left\{t_{p,s}:s\in S\right\}} \tag{4.17}$$

A ratio of 1 means that a method performs the best on that problem. A higher ratio means it needs more iterations or time compared to the best method. Finally, the performance profile function is defined as:

$$\rho_s(\tau) = \frac{size\left\{ p \in P : r_{p,t} \le t \right\}}{size\left\{ p \in P \right\}} \tag{4.18}$$

The performance profile represents the proportion of problems for which method s performs within a factor τ of the best-performing method. In the graph, the *x*-axis shows the performance ratio τ on a logarithmic scale, while the y-axis represents the fraction of problems satisfying this condition. When $\tau = 1$, the corresponding *y*-value indicates the proportion of problems for which a method achieves the best performance. As τ increases, the *y*-value approaches the overall fraction of problems successfully solved by the method, reflecting its robustness. Therefore, a method with a curve that rises rapidly and remains above the others is considered more efficient and robust than the competing methods (Dolan & Moré, 2001).

Performance profiles provide a graphical comparison of optimization methods using relative performance rather than raw average values. This approach reduces the influence of extreme results and includes unsuccessful runs in the evaluation, making performance profiles a reliable tool for benchmarking optimization algorithms (Dolan & Moré, 2001).

4.3 Fluence Map Optimization in IMRT

IMRT treatment planning consists of several optimization stages, each with a different objective. The main optimization problems include beam angle optimization, fluence map optimization, and leaf sequencing or direct aperture optimization (Ehrgott et al., 2008). Beam angle optimization determines the number and directions of the radiation beams, fluence map optimization determines the intensity of each beamlet for a fixed beam arrangement, and leaf sequencing converts the optimized fluence maps into deliverable multileaf collimator (MLC) shapes. These stages are usually solved separately because each stage has different optimization characteristics.

Among these stages, this study focuses on fluence map optimization because it has the greatest influence on the final dose distribution delivered to the patient. In FMO, each radiation beam is divided into many small beamlets, and the intensity of each beamlet is optimized so that the delivered dose matches the prescribed clinical dose as closely as possible while limiting the dose to the OARs. Since the desired dose

distribution is specified before the beamlet intensities are determined, fluence map optimization is formulated as an inverse treatment planning problem (Cao et al., 2022).

The relationship between the beamlet intensities and the delivered dose is represented by a dose-influence matrix, which describes the contribution of each beamlet to the dose received by every voxel. The dose distribution is obtained by multiplying the dose-influence matrix by the beamlet intensity vector (Cao et al., 2022). A typical IMRT treatment plan contains thousands of beamlets, resulting in a large-scale optimization problem with a large and often ill-conditioned dose-influence matrix (Maass et al., 2022). Consequently, an efficient optimization algorithm is required. The TTCG methods are suitable for solving this type of problem because they require low memory storage and are designed for large-scale unconstrained optimization problems.

Fluence map optimization is selected in this study for three main reasons. First, it is the stage that directly determines the quality of the dose distribution and is widely used as a benchmark for evaluating optimization algorithms in IMRT (Maass et al., 2022). Second, fluence map optimization is formulated as a large-scale continuous optimization problem, which matches the characteristics of the TTCG methods. Third, an openly available and validated IMRT optimization framework with a clinical prostate dataset from Maass et al. (2022) provides a realistic platform for evaluating the performance of the TTCG methods.

In this study, the fluence map optimization problem is applied to a prostate cancer case. A uniform dose is prescribed to the Planning Target Volume (PTV), while the radiation dose delivered to the rectum, which is an OAR, is restricted using a dose-volume constraint. This constraint specifies that only a limited percentage of the rectum is allowed to receive a dose higher than a prescribed threshold (Dumela, 2024; Maass et al., 2022). Therefore, the optimization aims to deliver the prescribed dose to the prostate while minimizing the amount of rectum tissue receiving excessive radiation. The quality of the resulting treatment plan is evaluated using the dose distribution, dose-volume histogram (DVH), and dose-volume metrics, which are presented in the following sections.

4.3.1 Fluence Map Optimization Formulation for IMRT

IMRT treatment planning is carried out through inverse planning. Instead of fixing the beam settings first and then computing the dose, inverse planning starts from the desired dose and works backwards to find the beamlet intensities that produce it (Ólafsson et al., 2005). This makes IMRT planning an optimization problem, where the beamlet intensities are the decision variables and the treatment goals are written as an objective function.

The link between the beamlet intensities and the delivered dose is linear. Each beam is divided into small beamlets, and the intensity of every beamlet is collected into a vector x. The dose received by the voxels of a structure is then obtained by multiplying a dose-influence matrix A by the intensity vector, so that the dose become Ax (Cao et al., 2022; Maass et al., 2022). Each structure, such as the tumour or an OAR, has its own dose-influence matrix. The linear relationship between the dose and x allows the difference between the delivered dose, Ax and the prescribed dose, d to be measured directly. As a result, the planning goal is to make Ax as close as possible to d.

The planning goals are turned into a single objective function using a least-squares penalty. For the tumour, the aim is to deliver the prescribed dose, so the term penalizes the squared difference between the delivered dose and the prescription. For an OAR, the aim is to keep the dose low, so a similar squared penalty is applied to the dose above the allowed level. Each term is given a weight that controls its relative importance, and the weighted terms are added together to form one objective function (Maass et al., 2022). As a result, the goals of delivering enough dose to the tumour and reducing the dose to healthy tissue are combined into a single objective function to be minimized.

The only physical requirement on the variables is that beamlet intensities cannot be negative, since a beam cannot deliver a negative dose. This is a simple bound, $x \geq 0$, and it is handled by projection during the optimization rather than by a general constraint solver. At every iteration, any negative intensity is set back to zero, which keeps the iterates physically deliverable. Since this bound is enforced using projection, the problem can be treated as an unconstrained minimization of the least-squares objective instead of using a constrained optimization method (Dumela, 2024; Maass et

al., 2022). As a result, the TTCG methods can be applied to the IMRT fluence map optimization problem because it is designed for unconstrained optimization.

A more challenging clinical requirement is the dose-volume constraint for the OAR, which states that only a certain percentage of the organ may receive a dose above a given limit. This constraint is nonconvex and cannot be written as a simple penalty term. Following Maass et al. (2022), an auxiliary variable is introduced to represent the allowable dose pattern in the OAR. The optimization then alternates between two steps, updating this auxiliary variable through a projection step, and minimizing the least-squares objective with respect to the beamlet intensities. The second step is an unconstrained subproblem solved using the TTCG method. In this way, the original IMRT problem is turned into a sequence of unconstrained least-squares problems, and the TTCG methods studied in this thesis are used to solve each subproblem. The detailed mathematical formulation of this process is presented in Section 5.2.2.

4.3.2 Dose-Volume Histogram (DVH)

The DVH is a common tool used to evaluate the quality of an IMRT treatment plan (Cao et al., 2022). A DVH shows the relationship between radiation dose and the volume of a structure receiving that dose. The horizontal axis represents the dose, while the vertical axis represents the percentage of volume. Each point on the DVH curve indicates the percentage of the structure that receives at least a certain dose value (Cao et al., 2022).

The shape of the DVH curve provides information about plan quality. For the PTV, an ideal DVH curve stays close to 100% volume until it reaches the prescribed dose and then drops sharply. This indicates that most of the target receives the required dose. For an OAR such as the rectum, the DVH curve should remain close to the origin, showing that only a small volume receives a high dose (Cao et al., 2022; Liu et al., 2021). By comparing the DVH curves before and after optimization, the effectiveness of the optimization process can be evaluated in terms of target coverage and healthy tissue protection.

4.3.3 Dose-Volume Metrics

In addition to the DVH, specific dose-volume metrics are used to evaluate the quality of an IMRT treatment plan. One of the most used metrics is D95, which represents the dose received by 95% of the PTV. D95 is widely used because it measures how well the prescribed dose covers the target volume. A D95 value that is close to the prescribed dose indicates that almost the entire tumour receives the intended radiation dose, reflecting good target coverage while reducing the risk of underdosing part of the tumour.

The quality of OAR protection is evaluated using dose-volume constraints. These constraints limit the percentage of an organ that is allowed to receive a dose above a specified value (Maass et al., 2022). For the prostate case considered in this study, target coverage is evaluated using the PTV D95 value, while rectum protection is evaluated by measuring the percentage of the rectum volume receiving a dose greater than 30 Gy.

In addition to treatment plan quality, the computational performance of the TTCG methods is evaluated using the NOI and CPU time, as introduced in Section 3.5. Together, these performance measures provide a comprehensive evaluation of both the optimization efficiency and the quality of the resulting IMRT treatment plan, which are discussed in Chapter 5.

4.4 Conclusion

This chapter has presented the implementation of the proposed TTCG methods used in this study. The numerical test setup was first introduced by presenting the fifteen standard unconstrained optimization test functions together with their initial points and numbers of variables. The performance profile was then described as the evaluation tool for comparing the efficiency and robustness of the TTCG methods in terms of the NOI and CPU time. Next, the IMRT fluence map optimization problem was formulated, including the mathematical model, the optimization framework, and the clinical prostate case considered in this study. Finally, the DVH and dose volume metrics were introduced as the criteria for evaluating the quality of the treatment plans. These implementations provide the foundation for the numerical experiments and the analysis of the TTCG methods presented in the following chapter.

CHAPTER 5
RESULTS AND DISCUSSIONS

5.1 Introduction

This chapter presents the results and discussion of this study. First, the numerical performance of the five TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR, and TTSFA, is evaluated using performance profiles based on the NOI and CPU time. The success rate of each TTCG method is also reported to assess its robustness. Based on these numerical results, the TTCG methods are compared to identify the method that provides the best balance between efficiency and robustness for solving standard unconstrained optimization problems.

Next, the five TTCG methods are implemented in a prostate IMRT fluence map optimization problem. The optimization performance is evaluated using the NOI, CPU time, and final objective value, while the resulting treatment plans are assessed using the dose distribution, DVH, and dose-volume metrics. Finally, the main findings from the numerical experiments and the IMRT application are discussed and summarized.

5.2 Results

This section presents the results in two stages. Section 5.2.1 presents the numerical comparison of the five TTCG methods using standard unconstrained optimization test functions. Based on these results, the TTCG methods are evaluated in terms of efficiency and robustness. Section 5.2.2 presents the implementation of the five TTCG methods in a prostate IMRT fluence map optimization problem, followed by a comparison of their optimization performance and treatment plan quality.

5.2.1 Numerical Result

The numerical results for TTRMIL, TTLAMR, TTKMAR, TTSMAR and TTSFA are shown in this section using performance profiles. Figures 5.1 and 5.2 show the performance profiles for NOI and CPU time, respectively.

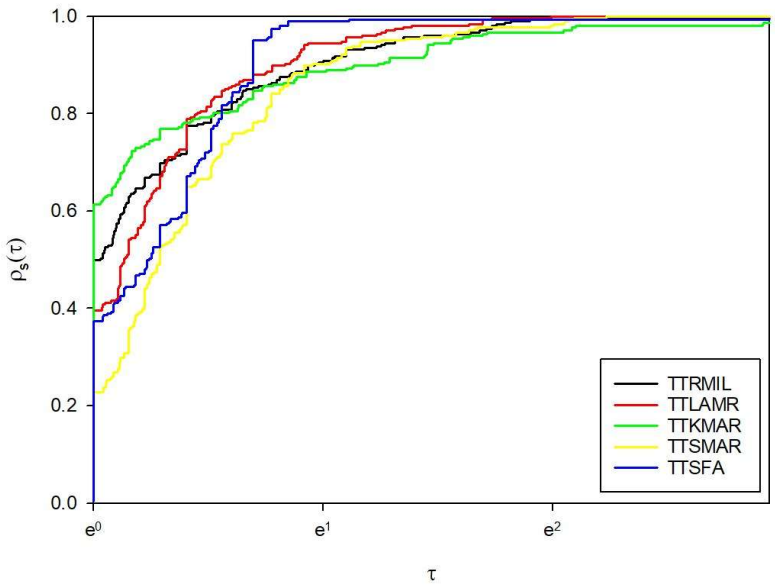


Figure 5.1 The Performance Profile of NOI

Figure 5.1 presents the performance profile of the five TTCG methods based on the NOI. The performance profile compares the efficiency of the TTCG methods by showing the proportion of test problems solved with the fewest iterations. A method with a curve that rises more rapidly and remains above the others is considered more efficient.

Based on Figure 5.1, TTKMAR achieves the highest curve near $\tau = 1$, indicating that it requires the fewest iterations for the largest proportion of the fifteen test functions. TTRMIL ranks second, followed by TTLAMR, TTSFA, and TTSMAR. As the performance ratio increases, the curves of all five TTCG methods gradually approach one, showing that each method can solve almost all the test problems. Therefore, TTKMAR is identified as the most efficient TTCG method in terms of the NOI.

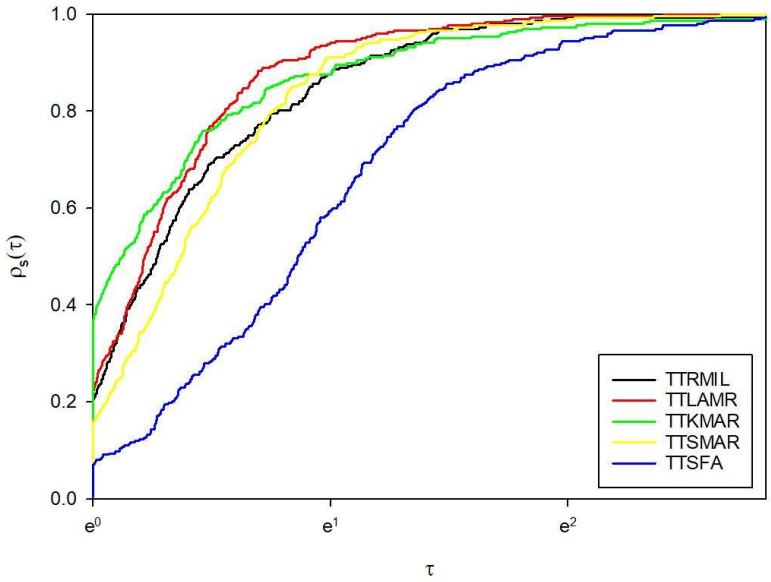


Figure 5.2 The Performance Profile of CPU Time

Figure 5.2 presents the performance profile of the five TTCG methods based on CPU time. The performance profile compares the computational efficiency of the TTCG methods by showing the proportion of test problems solved within the shortest computation time. A method with a higher curve near $\tau = 1$ is considered more computationally efficient. From Figure 5.2, TTKMAR again achieves the highest curve near $\tau = 1$, indicating that it requires the least CPU time for the largest proportion of the fifteen test functions. TTLAMR ranks second, followed by TTRMIL, TTSMAR, and TTSFA. As the performance ratio increases, the curves of all five TTCG methods converge to nearly one, indicating that all methods can solve almost all the test problems successfully. Therefore, TTKMAR demonstrates the best computational efficiency in terms of CPU time.

Since the performance profiles evaluate only efficiency, the overall selection of the TTCG method also considers robustness. The robustness of the five TTCG methods is evaluated based on the overall success rate, which is presented in Table 5.1. The success rate indicates the percentage of test problems successfully solved by each TTCG method. The results in Table 5.1 are discussed in the following section to identify the TTCG method that provides the best overall balance between efficiency and robustness. Additionally, these curves can be calculated manually using Equations (4.16), (4.17) and (4.18). The example of calculation is shown below:

a) By using Equation (4.16), find the minimum CPU time among all five TTCG methods:

$$\min\left\{t_{p,s} : s \in S\right\}$$
$$= \min\left\{0.0381,\ 0.0254,\ 0.0156,\ 0.0162, 0.0471\right\} = 0.0156$$

b) Calculate the performance ratio using Equation (4.17):

$$r_{p,s} = \frac{t_{p,s}}{\min\left\{t_{p,s} : s \in S\right\}}$$

$$r_p, TTRMIL = \frac{0.0381}{0.0156} = 2.4423$$
$$r_p, TTLAMR = \frac{0.0254}{0.0156} = 1.6282$$
$$r_p, TTKMAR = \frac{0.0156}{0.0156} = 1.000$$
$$r_p, TTSMAR = \frac{0.0162}{0.0156} = 2.4423$$
$$r_p, TTSFA = \frac{0.0471}{0.0156} = 2.4423$$

c) By using Equation (4.18), calculate the probability using the performance profile:

$$\rho_s(\tau) = \frac{size\left\{p \in P : r_{p,t} \le t\right\}}{size\left\{p \in P\right\}}$$

d) Calculate the percentage:

$$TTLAMR : \rho_S(\tau) = \frac{308}{308} \times 100\% = 100\%$$
$$TTSMAR : \rho_S(\tau) = \frac{308}{308} \times 100\% = 100\%$$
$$TTRMIL : \rho_S(\tau) = \frac{307}{308} \times 100\% = 99.68\%$$
$$TTSFA : \rho_S(\tau) = \frac{306}{308} \times 100\% = 99.35\%$$
$$TTKMAR : \rho_S(\tau) = \frac{305}{308} \times 100\% = 99.03\%$$

Based on all the results, TTKMAR is the most efficient method as it uses the fewest iterations and least CPU time. However, TTLAMR and TTSMAR are the most robust, each achieving a 100% success rate, while TTSFA solves 99.35% of the problems. Overall, TTLAMR provides the best combination of efficiency and robustness, because it attains the full 100% success rate while remaining competitive in both NOI and CPU time. This makes it the most suitable coefficient for solving large-scale problems such as IMRT optimization. Based on both figures above, the results of the right side of the graph are tabulated in Table 5.1.

Table 5.1

The Success Rate of TTCG Methods

TTCG	Success Rate
TTRMIL	99.68%
TTLAMR	100%
TTKMAR	99.03%
TTSMAR	100%
TTSFA	99.35%

From the success rates presented in Table 5.1, TTLAMR and TTSMAR successfully solve all the test problems, each achieving a success rate of 100%. In contrast, TTRMIL, TTSFA, and TTKMAR solve most of the test problems but fail on a few cases, resulting in success rates below 100%. These results indicate that TTLAMR and TTSMAR are the most robust TTCG methods because both methods successfully solve every test problem.

Considering all the numerical results, TTKMAR is the most efficient TTCG method because it requires the fewest iterations and the least CPU time. However, TTKMAR does not successfully solve every test problem, achieving a success rate of only 99.03%. Although TTLAMR and TTSMAR require slightly more iterations and CPU time than TTKMAR, both methods achieve a 100% success rate. In this study, the best balance is defined as a TTCG method that demonstrates both high efficiency and

complete robustness. TTLAMR satisfies these two criteria because it achieves a 100% success rate while maintaining performance that is very close to TTKMAR in terms of the NOI and CPU time. Although TTSMAR also achieves a 100% success rate, its performance profile rises more slowly than that of TTLAMR, indicating lower computational efficiency. Therefore, TTLAMR provides the best overall balance between efficiency and robustness among the five TTCG methods and is selected for application to the IMRT fluence map optimization problem in the next section.

5.2.2 Application

Before presenting the numerical results, the mathematical formulation of the IMRT fluence map optimization problem is introduced. The formulation is presented progressively, beginning with the general dose-fitting model and ending with the specific non-negative least-squares problem considered in this study.

In general, fluence map optimization seeks the beamlet intensity vector x that makes the delivered dose match the clinical goals for every structure. This objective function is written as

$$\min_{x\geq 0} \frac{1}{2}\|Ax-d\|^2 \tag{5.1}$$

Here, A is the dose-influence matrix, x is the vector of beamlet intensities, and d is the prescribed dose, so that Ax is the delivered dose.

However, in a clinical treatment plan, multiple structures must be considered simultaneously, including the tumour and surrounding OAR. Therefore, the general dose-fitting model in Equation (5.1) is extended by combining the dose mismatch for all structures into a weighted objective function, given by

$$F(x)=\sum_s \frac{c_s}{2}\|A_s x-d_s\|^2 \tag{5.2}$$

For each structure s, A_s is the corresponding dose-influence matrix, d_s is the prescribed or desired dose, and c_s is the weight assigned to the structure. The term $A_s x$ therefore represents the delivered dose to structure s, while the squared error measures

the difference between the delivered and desired doses. The overall objective in Equation (5.2) combines these errors across all structures.

Within this formulation, the tumour term encourages the delivered dose to approach the prescribed tumour dose, while the OAR terms penalize doses that exceed the allowed levels. The beamlet intensities are then determined by minimizing the objective function subject to the non-negativity constraint, which is written as

$$\min_{x \geq 0} F(x) \tag{5.3}$$

This formulation determines the beamlet intensities that best balance tumour coverage and healthy tissue sparing (Maass et al., 2022).

For the prostate case in this study, there are two structures. The PTV is prescribed a uniform dose, while the rectum is treated as the OAR. Since the dose volume constraint is nonconvex, it cannot be included directly in the objective function, $F(x)$. Following Maass et al. (2022), it is imposed through an auxiliary variable w that represents the target dose pattern in the rectum, and the problem is written as the non-negative least-squares problem in Equation (5.4). In this formulation, the factors $2n_t$ and $2n_r$ normalize each term by twice the number of tumour and rectum voxels, respectively. Here, α is the weighting parameter that balances the tumour and rectum objectives. The auxiliary variable w represents the allowable dose distribution in the rectum. The constraint limits the number of rectum voxels that are allowed to receive a dose above the prescribed limit, with the sparsity of $w$representing the dose-volume constraint. The final term is a small Tikhonov regularization term controlled by the parameter λ, which is included to improve numerical stability.

The selected TTLAMR method is implemented for a clinical prostate IMRT case using the fluence map optimization framework proposed by Maass et al. (2022), with the CORT prostate dataset. The case consists of one PTV, the prostate, prescribed a uniform dose of 81 Gy, and one OAR, the rectum, constrained so that no more than 30% of its volume receives more than 30 Gy. The treatment uses seven coplanar beams positioned at gantry angles of 0, 52, 104, 156, 208, 260, and 312 degrees. The optimization variables are the beamlet intensities that define the fluence map for each beam.

The fluence map optimization problem solved in this study follows the formulation of Maass et al. (2022). The objective is to deliver the prescribed dose to the tumour while limiting the dose to the rectum. Since the dose-volume constraint is nonconvex, it is imposed on an auxiliary variable w, and the optimization problem can be written as the following non-negative least-squares problem

$$\min_{x\geq 0, w} \frac{1}{2n_t}\left\|A_t x - d_t\right\|_2^2 + \frac{a}{2n_r}\left\|w - (A_r x - d_r)\right\|_2^2 + \frac{\lambda}{2}\left\|x\right\|_2^2 \tag{5.4}$$

such that $\left\|w_+\right\|_0 \leq k$, where x is the vector beamlet intensities. The matrices A_t and A_r represent the dose-influence matrices for the tumour and rectum, respectively, while d_t and d_r represent the prescribed doses. In this study, the optimization involves 986 beamlet intensities from seven treatment beams, corresponding to the active beamlets of the CORT prostate dataset after the beamlet grid is restricted to those that intersect the patient volume. The prescribed tumour dose is 81 Gy for 6770 prostate voxels, while the rectum dose limit is 30 Gy for 1764 rectum voxels. The parameter α is set to 1 to balance the tumour and rectum objectives, and $\lambda = 1\times 10^{-8}$ is included to improve numerical stability. This parameter is the coefficient of the regularization term in Equation (5.4).

This problem is solved by alternating between a projected update of the auxiliary variable w and a minimization over the beamlet intensities x. For a fixed w, the update of x is obtained by solving the following bound-constrained quadratic problem:

$$x = \arg\min_{x\geq 0} \frac{1}{2}\left\|Ax - d(w)\right\|_2^2 = \arg\min_{x\geq 0}\left(\frac{1}{2}x^T Hx + f^T x\right) \tag{5.5}$$

where $H = A^T A$ is the Hessian matrix and $f = -A^T d(w)$ is the linear term. Here H is fixed across all outer iterations because the dose-influence matrices do not change, whereas the linear term f depends on the current auxiliary variable w and is therefore updated at each outer iteration. Each TTCG method is then applied to solve this quadratic subproblem. Since H is constant, the exact line search formula presented in Chapter 3 can be used directly. This allows each TTCG method to be applied to the IMRT fluence map optimization problem.

To evaluate the performance of the five TTCG methods on a clinical treatment planning problem, each TTCG method is implemented in the IMRT fluence map optimization framework. Only the optimization algorithm is replaced, while the dose calculation, DVH generation, beamlet visualization, and treatment evaluation procedures remain unchanged. Each TTCG method is implemented in turn, with the corresponding search direction defined by the CG coefficient described in Chapter 3. All five TTCG methods use the same problem formulation, exact line search strategy, initial point, and stopping criterion to ensure a fair comparison. Consequently, any differences in performance are attributed solely to the TTCG method employed. Since beamlet intensities cannot be negative in practice, the non-negativity constraint is enforced throughout the optimization process. This implementation enables the computational performance of the TTCG methods to be evaluated in the context of IMRT fluence map optimization.

The performance of the selected TTLAMR method on the prostate IMRT case is presented in Table 5.2. The optimization performance is evaluated using the NOI, CPU time, and the final objective value. Among these measures, the final objective value is used to assess the quality of the optimized solution, while NOI and CPU time indicate the computational efficiency of the optimization process. The resulting treatment plan is further evaluated using the percentage of rectum volume receiving a dose greater than 30 Gy.

Table 5.2

Performance of the Five TTCG Methods on the Prostate IMRT Case

Method	**NOI**	**CPU Time (s)**	**Final Objective Value**	**PTV D95 (Gy)**	**Rectum volume above 30 Gy (%)**
TTRMIL	47	165.12	5.5718	79.20	34.28
TTLAMR	46	153.98	5.5709	79.19	34.47
TTKMAR	40	126.07	5.5810	79.19	34.53
TTSMAR	43	150.86	5.5813	79.19	34.16
TTSFA	62	207.93	5.6353	79.29	37.80

Table 5.2 compares the five TTCG methods on the prostate IMRT fluence map optimization problem. In terms of NOI, TTKMAR is the most efficient method, converging in 40 iterations, followed by TTSMAR with 43 iterations and TTLAMR with 46 iterations. TTRMIL is close behind with 47 iterations, while TTSFA is the least efficient, requiring 62 iterations. The CPU times follow the same order for the fastest methods, with TTKMAR recording the lowest CPU time. This ordering is consistent with the test-function study, where TTKMAR is also identified as the most efficient coefficient.

In terms of solution quality, TTLAMR reaches the lowest final objective value of 5.5709, meaning it converges to the best fluence map among the five methods even though it needs a few more iterations than TTKMAR to do so. TTRMIL, TTKMAR, and TTSMAR reach very similar objective values between 5.5718 and 5.5813, while TTSFA gives the highest objective value of 5.6353, indicating a less accurate solution.

The clinical results show the same pattern. The four leading methods, TTKMAR, TTSMAR, TTLAMR, and TTRMIL, all achieve a PTV D95 of about 79.19 to 79.20 Gy, which is close to the prescribed dose of 81 Gy and indicates good target coverage. Their rectum values are also similar, ranging from 34.16% to 34.53% of the rectum volume receiving more than 30 Gy, with TTSMAR giving the best rectum sparing at 34.16%. TTSFA again performs worst, with a higher rectum value of 37.80%, meaning more of the rectum receives a high dose. Its objective value also rose slightly in the later iterations before the outer loop stopped, which shows that TTSFA is the least stable of the five coefficients on this problem.

Overall, the five TTCG methods can be divided into two groups. TTKMAR, TTLAMR, TTRMIL, and TTSMAR produce clinically comparable treatment plans, while TTSFA performs less favourably in terms of both computational efficiency and treatment plan quality. Although no single TTCG method performs best for every evaluation measure, the differences among the four leading methods are relatively small.

Among these methods, TTKMAR requires the fewest iterations, making it the most computationally efficient method. TTLAMR achieves the lowest final objective value, whereas TTSMAR provides the smallest percentage of rectum volume receiving a dose greater than 30 Gy. However, all four leading TTCG methods produce very similar treatment plans, with a PTV D95 of approximately 79.19 Gy and about

34% of the rectum volume receiving a dose greater than 30 Gy. These results indicate that the treatment plans produced by the four methods are clinically comparable.

Since the treatment plan quality is almost identical, computational efficiency becomes an important consideration. In this study, the NOI is used as the main indicator of computational efficiency because it depends only on the optimization algorithm, whereas CPU time may also be influenced by the computing environment. From this perspective, TTKMAR demonstrates the best computational performance by producing a clinically comparable treatment plan using the fewest iterations. Although TTLAMR achieves the lowest final objective value, this improvement does not result in a noticeable enhancement in the treatment plan quality. Therefore, the lower objective value represents an improvement in the optimization result rather than a clinically significant advantage.

These findings indicate that the performance ranking obtained from the standard unconstrained optimization test functions does not transfer directly to the IMRT application. While TTLAMR provides the best balance between efficiency and robustness on the standard test functions, TTKMAR demonstrates the highest computational efficiency for the IMRT fluence map optimization problem without compromising treatment plan quality.

To further evaluate the treatment plan, the dose distribution, DVH, and optimized fluence maps are presented in Figures 5.3 to 5.5. Since the treatment plans produced by the four leading TTCG methods are clinically comparable, the results for the TTLAMR method are presented as a representative example. These figures illustrate the quality of the optimized treatment plan rather than comparing the performance of the TTCG methods, which has already been evaluated using the quantitative measures presented in Table 5.2.

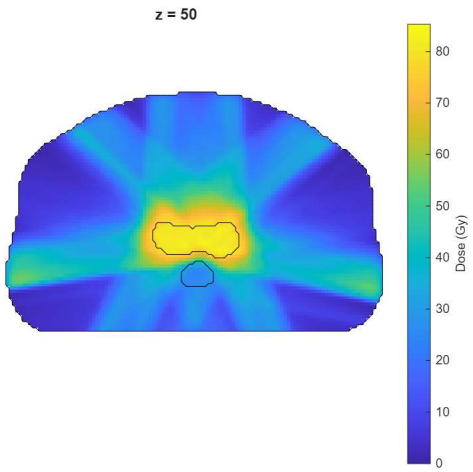


Figure 5.3 Dose Distribution Obtained Using the TTLAMR Method

The resulting dose distribution is shown in Figure 5.3. The figure shows an axial slice of the patient at z = 50, which passes through the centre of the prostate target. The colour scale on the right shows the dose in Gray (Gy), where dark blue represents a low dose close to 0 Gy and yellow represents a high dose of about 80 Gy, which is near the prescribed dose of 81 Gy.

The high-dose region, shown in yellow, is concentrated on the prostate target, which is marked by the black contour in the middle of the slice. The shape of the yellow region closely follows the shape of the target contour, which shows that the dose is conformal to the PTV. The rectum contour, located immediately below the target, receives a much lower dose, shown by the cooler blue and green shades inside it. This is important because the rectum is the main OAR in this prostate case, and its dose must be kept low to avoid damage to healthy tissue.

The seven beam directions are also visible in the figure as faint paths of light blue and green that spread outward from the target toward the body surface. Each path represents the entrance dose of one beam. Individually, each beam carries only a moderate dose, but the beams overlap at the target, so the dose adds up only at the PTV. This overlapping produces the desired dose concentration on the target while the surrounding healthy tissue, which is crossed by only one beam at a time, receives a much lower dose. In addition, the dose drops quickly outside the target region, which shows a steep dose gradient between the PTV and the nearby normal tissue. Overall, Figure 5.3 shows that the TTLAMR method produces a dose distribution that covers the target well while sparing the rectum and the surrounding healthy tissue.

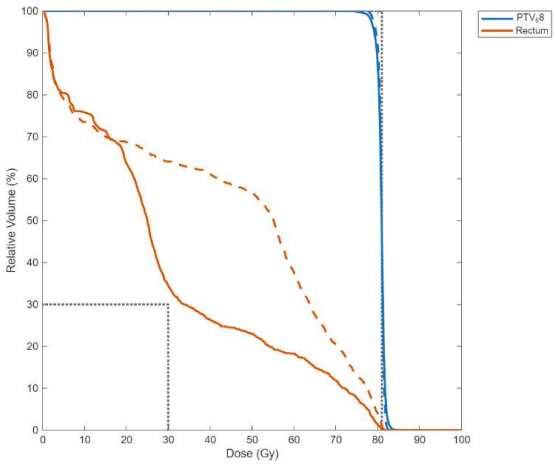


Figure 5.4 Dose-Volume Histogram for the PTV and Rectum (Solid: Optimized, Dashed: Initial)

The DVH in Figure 5.4 is used to evaluate the quality of the treatment plan. The horizontal axis shows the dose in Gray (Gy), while the vertical axis shows the percentage of the structure volume that receives at least that dose. The solid lines represent the optimized plan, while the dashed lines represent the initial plan.

For the PTV, shown in blue, the curve remains close to 100% of the volume before dropping steeply near the prescribed dose of 81 Gy. This steep fall shows that the dose inside the target is uniform. The PTV D95 value is 79.19 Gy, which means 95% of the target volume receives at least 79.19 Gy. This value is close to the prescribed dose and shows good target coverage.

For the rectum, shown in orange, the optimized curve lies well below the initial curve over most of the dose range, which means a smaller volume of the rectum receives each dose level after optimization. The dotted lines mark the rectum constraint, where no more than 30% of the volume should receive more than 30 Gy. In the optimized plan, 34.47% of the rectum volume receives more than 30 Gy, which is slightly above the limit. Although the constraint is not fully satisfied, the rectum dose is clearly reduced compared to the initial plan.

Overall, the DVH shows that the TTLAMR method maintains good target coverage while reducing the dose to the rectum, confirming that the method can produce a clinically acceptable treatment plan.

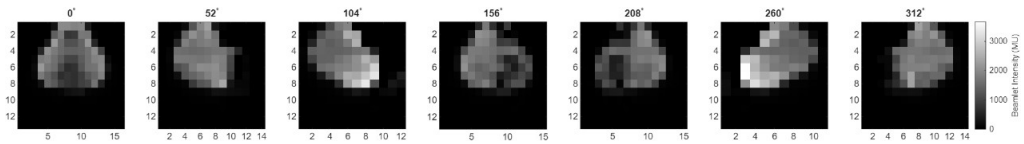


Figure 5.5 Optimized Fluence Maps for the Seven Treatment Beams

Finally, the optimized fluence maps for the seven treatment beams are shown in Figure 5.5. Each map represents the beamlet intensities of one beam, obtained from the TTLAMR optimization, and each belongs to a different gantry angle spaced evenly around the patient. Within each map, every pixel corresponds to one beamlet, and brighter regions mean higher beamlet intensities.

The bright area in each map roughly follows the shape of the prostate target as seen from that beam direction. This is expected, since the beamlets that pass through the PTV are given higher intensities to deliver the prescribed dose, while the beamlets that pass through the rectum or other healthy tissue are given lower intensities. The intensity pattern also changes from one gantry angle to another, because the shapes and positions of the target and the rectum are different from each viewing direction. This non-uniform pattern is the main feature of IMRT, where the intensity within each beam is modulated instead of being kept constant.

These fluence maps are the final output of the fluence map optimization process. Together, they form the optimized beamlet intensity distributions used in the IMRT treatment plan, and they produce the dose distribution and DVH results shown in Figures 5.3 and 5.4.

5.3 Conclusion

This chapter presents the application of the five TTCG methods to a clinical prostate IMRT fluence map optimization problem. Each TTCG method is implemented by replacing the original optimization solver while maintaining the same IMRT framework, exact line search strategy, and stopping criterion. The results show that the four leading TTCG methods, namely TTRMIL, TTLAMR, TTKMAR, and TTSMAR, produce clinically comparable treatment plans, achieving a PTV D95 of approximately 79.19 Gy for the prescribed dose of 81 Gy and limiting the rectum volume receiving more than 30 Gy to between 34.16% and 34.53%.

Although no single TTCG method performs best for all evaluation measures, each method demonstrates its own strength. TTKMAR requires the fewest iterations, TTLAMR achieves the lowest final objective value, and TTSMAR provides the greatest rectum sparing, whereas TTSFA performs less favourably in terms of both computational efficiency and treatment plan quality. These results indicate that the choice of TTCG method depends on the performance measure being considered.

The dose distribution, DVH, and optimized fluence maps further confirm that the four leading TTCG methods produce clinically comparable treatment plans. Overall, the results demonstrate that the TTCG methods are applicable to IMRT fluence map optimization and can generate clinically acceptable treatment plans. In addition, the findings show that the performance ranking obtained from the standard unconstrained optimization test functions does not necessarily carry over to the IMRT application.

CHAPTER 6
CONCLUSION AND RECOMMENDATION

6.1 Introduction

This final chapter concludes the research. The main findings are first summarized based on the three research objectives in Chapter 1, covering the numerical comparison of the TTCG methods and their application to IMRT fluence map optimization. Several recommendations for future research are then given.

6.2 Conclusion

This study investigates the performance of the TTCG method under exact line search and its applicability to IMRT fluence map optimization. The study is based on three research objectives, and each objective has been successfully achieved.

The first objective is to solve standard optimization test functions using the TTCG method. This is achieved by implementing five TTCG coefficients, namely TTRMIL, TTLAMR, TTKMAR, TTSMAR and TTSFA, in MATLAB and applying them to solve 15 standard test functions from the CUTE library. The functions are tested with various numbers of variables and initial points using an exact line search, and the sufficient descent property is verified numerically. Solving these test problems successfully shows that the implementation works correctly.

The second objective is to compute and analyze the performance of the TTCG method in terms of the NOI and CPU time. Using the performance profile, the five coefficients are compared on both measures. TTKMAR is found to be the most efficient method, while TTLAMR and TTSMAR are the most robust, each solving all test problems with a 100% success rate. Taking efficiency and robustness together, TTLAMR is identified as the coefficient providing the best overall balance.

The third objective is to implement the TTCG methods in an IMRT fluence map optimization problem. All five coefficients are applied to a clinical prostate IMRT case using the fluence map optimization framework of Maass et al. (2022), with the same setup, exact line search, and stopping criterion for every method. The four leading coefficients, TTKMAR, TTSMAR, TTLAMR, and TTRMIL, all produce clinically

comparable treatment plans, achieving a PTV D95 of about 79.19 Gy against the prescribed 81 Gy and limiting the rectum volume above 30 Gy to between 34.16% and 34.53%. In terms of efficiency, TTKMAR converges in the fewest iterations (40), while TTLAMR attains the lowest final objective value (5.5709). TTSFA performs worst on both efficiency and plan quality. These results show that the TTCG method can be applied successfully to IMRT fluence map optimization, and that several of the coefficients are suitable for this purpose.

In summary, the TTCG method provides a reliable approach for solving both unconstrained optimization problems and large-scale IMRT fluence map optimization. The findings show that the best performing coefficient on standard test functions is not necessarily the best for clinical application. Although TTLAMR provides the best balance of efficiency and robustness on the standard test functions, different coefficients perform best under different evaluation measures in the IMRT application. TTKMAR converges in the fewest iterations, TTLAMR achieves the lowest final objective value, and TTSMAR provides the best rectum sparing. Since these three coefficients produce clinically equivalent treatment plans, the quality of the final fluence map becomes the deciding factor. Based on the final objective value, TTLAMR is identified as the most suitable coefficient for the IMRT application because it converges to the best fluence map, reaching the lowest objective value among the five methods while keeping the number of iterations low. These findings show that the selection of a TTCG coefficient for IMRT should be based on its performance in the clinical optimization problem rather than solely on standard optimization test functions. Overall, the consistent performance of TTLAMR, TTKMAR, and TTSMAR confirms that the TTCG method is an effective and suitable optimization approach for large-scale radiotherapy treatment planning.

6.3 Recommendation

Although the research objectives are met, several directions are recommended for future work to extend and strengthen the findings of this study.

First, this study uses an exact line search to provide a consistent benchmark for comparing the TTCG coefficients. Although this gives a fair comparison, exact line search is computationally expensive for large-scale problems such as IMRT. Future research may apply inexact line search techniques, such as Armijo backtracking or the strong Wolfe conditions, which usually require fewer function evaluations and may

reduce the overall computational cost while maintaining acceptable performance (Ayansiji, 2026; Dzailani et al., 2025).

Second, the finding that the best coefficient on the test functions is not the best in the IMRT application should be tested more widely. This study uses a single prostate case with one target and one OAR. Future research can apply the five coefficients to more clinical cases and to more complex treatment sites with several OARs and several dose-volume constraints, to determine whether the ranking observed here is consistent across different IMRT problems.

Third, since no single coefficient is best on every measure, future work may develop hybrid or scaled TTCG formulations that combine the efficiency of TTKMAR, the robustness of TTSMAR, and the solution accuracy of TTLAMR within a single method. Studying the theoretical convergence properties of such formulations, together with their behaviour on IMRT problems, could lead to a coefficient that performs well on both efficiency and plan quality.

Fourth, the weaker and less stable behaviour of TTSFA in the IMRT application suggests that the numerical stability of the coefficients deserves further study. Future work may examine why some coefficients lose accuracy on ill-conditioned IMRT problems and how safeguards or restart strategies can be designed to keep them stable, so that a wider range of TTCG coefficients can be used reliably in treatment planning.

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APPENDICES

APPENDIX 1

Numerical Result for Number of Iterations (NOI)

Functions	Variables	Initial Points	TTRMIL	TTLAMR	TTKMAR	TTSMAR	TTSFA
Extended Rosenbrock	2	(2,2)	28	37	71	15	22
		(5,5)	23	27	40	24	20
		(7,7)	19	19	46	25	24
		(9,9)	25	31	48	39	20
	4	(2,...,2)	28	37	71	15	22
		(5,...,5)	23	27	40	25	20
		(7,...,7)	19	19	46	25	24
		(9,...,9)	25	31	48	40	23
	10	(2,...,2)	31	37	73	17	22
		(5,...,5)	21	29	42	25	20
		(7,...,7)	19	23	48	25	24
		(9,...,9)	25	31	48	39	27

	100	(2,...,2)	31	37	73	17	22
		(5,...,5)	24	42	24	25	21
		(7,...,7)	19	22	48	25	24
		(9,...,9)	25	33	48	40	23
	500	(2,...,2)	33	37	73	17	22
		(5,...,5)	23	29	42	25	21
		(7,...,7)	19	19	48	25	24
		(9,...,9)	25	37	48	37	22
	1000	(2,...,2)	31	37	73	17	22
		(5,...,5)	23	29	42	24	21
		(7,...,7)	20	25	48	26	28
		(9,...,9)	24	35	28	37	22
	10000	(2,...,2)	31	40	75	17	25
		(5,...,5)	23	29	44	24	24
		(7,...,7)	22	21	48	26	25

		(9,...,9)	27	34	40	33	27
Extended White & Holst	2	(2,2)	19	23	69	23	29
		(5,5)	24	24	45	28	24
		(9,9)	40	18	145	52	30
		(11,11)	22	28	66	40	44
	4	(2,...,2)	19	23	69	24	29
		(5,...,5)	24	27	45	28	24
		(9,...,9)	40	18	145	53	30
		(11,...,11)	53	22	27	66	43
	10	(2,...,2)	19	23	69	24	34
		(5,...,5)	26	27	47	28	24
		(9,...,9)	49	18	98	53	30
		(11,...,11)	22	27	94	52	43
	100	(2,...,2)	20	23	69	24	30
		(5,...,5)	26	27	47	28	25

		(9,...,9)	39	18	147	43	34
		(11,...,11)	22	27	94	57	44
	500	(2,...,2)	20	29	71	24	20
		(5,...,5)	30	28	47	28	25
		(9,...,9)	47	20	103	56	30
		(11,...,11)	25	30	69	39	76
	1000	(2,...,2)	23	23	71	24	42
		(5,...,5)	30	27	47	28	26
		(9,...,9)	41	20	155	36	33
		(11,...,11)	24	30	102	49	51
Perturbed Quadratic	2	(3,3)	3	3	3	3	3
		(7,7)	3	3	3	3	4
		(9,9)	3	3	3	4	4
		(13,13)	3	3	3	4	4
	4	(3,...,3)	14	14	4	14	5

		(7,...,7)	15	15	5	15	5
		(9,...,9)	15	15	5	16	6
		(13,...,13)	15	15	5	16	6
	10	(3,...,3)	26	26	11	31	11
		(7,...,7)	28	27	11	33	11
		(9,...,9)	28	27	11	33	11
		(13,...,13)	28	28	11	34	11
Raydan 1	2	(3,3)	5	5	5	9	8
		(8,8)	6	7	6	13	11
		(13,13)	6	10	6	12	10
		(18,18)	27	28	35	52	7
	4	(3,...,3)	11	11	11	11	11
		(8,...,8)	16	12	15	17	18
		(13,...,13)	22	21	24	25	21
		(18,...,18)	27	28	35	52	18

	10	(3,...,3)	21	21	21	24	20
		(8,...,8)	38	37	27	46	31
		(13,...,13)	33	32	34	76	35
		(18,...,18)	27	28	35	52	34
Extended Tridiagonal 1	2	(2,2)	64	36	26	19	18
		(25,25)	3	3	3	3	3
		(35,35)	2	2	2	2	2
		(45,45)	2	2	2	2	2
	4	(2,...,2)	87	34	30	19	18
		(25,...,25)	3	3	3	3	3
		(35,...,35)	2	2	2	2	2
		(45,...,45)	2	2	2	2	2
	10	(2,...,2)	93	34	36	21	18
		(25,...,25)	3	3	3	3	3
		(35,...,35)	2	2	2	3	2

		(45,...,45)	2	2	2	2	2
	100	(2,...,2)	102	43	56	70	18
		(25,...,25)	26	16	148	10	8
		(35,...,35)	3	3	3	3	3
		(45,...,45)	3	3	3	3	3
	500	(2,...,2)	111	53	76	75	18
		(25,...,25)	38	19	166	48	9
		(35,...,35)	3	3	3	3	3
		(45,...,45)	3	3	3	3	3
	1000	(2,...,2)	156	98	86	140	18
		(25,...,25)	54	29	174	84	9
		(35,...,35)	3	3	3	3	3
		(45,...,45)	3	3	3	3	3
Diagonal 4	2	(5,5)	3	3	3	5	5
		(10,10)	3	3	3	5	5

		(20,20)	3	3	3	5	6
		(30,30)	4	4	4	5	6
	4	(5,...,5)	3	3	3	5	5
		(10,...,10)	3	3	3	5	5
		(20,...,20)	4	4	4	5	6
		(30,...,30)	4	4	4	5	6
	10	(5,...,5)	3	3	3	5	5
		(10,...,10)	3	3	3	5	6
		(20,...,20)	4	4	4	5	6
		(30,...,30)	4	4	4	5	6
	100	(5,...,5)	4	4	4	5	6
		(10,...,10)	4	4	4	5	6
		(20,...,20)	4	4	4	6	8
		(30,...,30)	4	4	4	6	8
	500	(5,...,5)	4	4	4	5	6

		(10,...,10)	4	4	4	6	8
		(20,...,20)	4	4	4	6	8
		(30,...,30)	4	4	4	6	8
	1000	(5,...,5)	4	4	4	6	6
		(10,...,10)	4	4	4	6	8
		(20,...,20)	4	4	4	6	8
		(30,...,30)	4	4	4	7	8
	10000	(5,...,5)	4	4	4	6	8
		(10,...,10)	4	4	4	7	8
		(20,...,20)	4	4	4	7	9
		(30,...,30)	5	5	4	7	9
Extended Himmelblau	2	(2,2)	7	7	6	7	10
		(6,6)	9	9	6	13	12
		(14,14)	9	9	6	13	12
		(24,24)	8	8	6	12	12

	4	(2,...,2)	8	7	6	7	10
		(6,...,6)	9	9	6	13	12
		(14,...,14)	9	9	6	13	12
		(24,…,24)	8	8	6	13	12
	10	(2,...,2)	8	7	6	7	10
		(6,...,6)	9	9	6	13	12
		(14,...,14)	9	9	6	14	12
		(24,…,24)	9	9	6	13	12
	100	(2,...,2)	8	8	6	7	10
		(6,...,6)	10	10	6	14	13
		(14,...,14)	9	9	6	14	13
		(24,…,24)	9	9	6	14	12
	500	(2,...,2)	9	8	6	7	11
		(6,...,6)	10	10	6	15	14
		(14,...,14)	9	9	6	15	13

		(24,…,24)	9	9	6	14	13
	1000	(2,...,2)	9	8	6	7	12
		(6,...,6)	10	10	6	15	14
		(14,...,14)	10	9	6	15	14
		(24,…,24)	9	9	6	15	13
	10000	(2,...,2)	9	9	7	7	12
		(6,...,6)	11	11	7	16	15
		(14,...,14)	10	10	7	16	14
		(24,…,24)	10	10	7	15	14
Extended DENSCHNB	2	(3,3)	6	6	6	7	9
		(8,8)	9	9	9	8	10
		(15,15)	8	9	8	17	13
		(21,21)	8	9	8	17	12
	4	(3,...,3)	6	6	6	7	9
		(8,...,8)	9	9	9	8	10

		(15,...,15)	8	9	8	17	13
		(21,…,21)	8	9	8	17	12
	10	(3,...,3)	6	6	6	7	9
		(8,...,8)	9	9	9	9	10
		(15,...,15)	8	9	8	17	14
		(21,…,21)	9	9	8	17	13
	100	(3,...,3)	7	7	6	8	9
		(8,...,8)	9	10	9	10	11
		(15,...,15)	8	9	8	18	14
		(21,…,21)	9	10	9	18	13
	500	(3,...,3)	7	7	6	8	9
		(8,...,8)	8	10	9	10	11
		(15,...,15)	8	10	8	18	14
		(21,…,21)	9	10	9	18	14
	1000	(3,...,3)	7	7	6	8	10

		(8,...,8)	9	10	10	10	12
		(15,...,15)	8	10	8	18	14
		(21,…,21)	9	10	9	18	14
	10000	(3,...,3)	7	7	7	8	11
		(8,...,8)	10	10	10	10	12
		(15,...,15)	9	10	9	19	14
		(21,…,21)	9	10	9	19	14
FLETCHCR	2	(5,5)	3	3	3	3	3
		(10,10)	4	4	4	4	4
		(15,15)	4	4	4	4	4
		(20,20)	4	4	4	5	4
	4	(5,...,5)	16	16	14	16	14
		(10,...,10)	16	17	17	22	15
		(15,...,15)	23	23	20	27	18
		(20,…,20)	22	22	20	22	13

	10	(5,...,5)	24	24	21	28	24
		(10,...,10)	32	32	30	40	29
		(15,...,15)	40	40	34	50	34
		(20,…,20)	41	46	36	59	35
Hager	2	(4,4)	5	5	4	7	7
		(8,8)	5	5	4	7	7
		(18,18)	6	6	6	5	6
		(21,21)	5	9	5	16	9
	4	(4,...,4)	8	8	8	9	9
		(8,...,8)	8	8	8	9	9
		(18,...,18)	10	10	9	8	8
		(21,…,21)	134	147	111	105	20
	10	(4,...,4)	12	12	12	13	12
		(8,...,8)	11	11	12	13	12
		(18,...,18)	15	15	14	11	11

		(21,…,21)	41	41	39	33	35
	100	(4,...,4)	26	26	25	28	26
		(8,...,8)	25	26	30	28	26
		(18,...,18)	31	39	30	35	26
		(21,…,21)	134	147	111	105	83
Extended Penalty	2	(3,3)	15	13	7	11	9
		(7,7)	12	9	8	9	7
		(10,10)	11	9	8	9	6
		(23,23)	8	6	7	11	9
	4	(3,...,3)	17	15	6	11	9
		(7,...,7)	16	15	6	11	9
		(10,...,10)	17	15	6	11	11
		(23,...,23)	8	8	8	12	9
	10	(3,...,3)	21	17	7	12	9
		(7,...,7)	20	17	7	9	11

		(10,...,10)	12	11	8	9	8
		(23,...,23)	21	17	8	12	12
	100	(3,...,3)	19	18	13	20	FAIL
		(7,...,7)	FAIL	15	FAIL	10	FAIL
		(10,...,10)	19	13	FAIL	10	19
		(23,...,23)	17	23	FAIL	10	14
Extended Maratos	2	(2,2)	19	22	25	18	17
		(5,5)	21	19	32	27	36
		(8,8)	25	27	25	24	39
		(11,11)	19	29	21	27	32
	4	(2,...,2)	19	19	27	18	17
		(5,...,5)	21	19	32	27	36
		(8,...,8)	25	23	27	24	42
		(11,...,11)	19	26	22	28	32
	10	(2,...,2)	19	31	25	18	17

		(5,...,5)	21	27	32	26	36
		(8,...,8)	25	26	23	27	42
		(11,...,11)	23	29	24	38	32
Sphere	2	(2,2)	1	1	1	1	1
		(8,8)	1	1	1	1	1
		(16,16)	1	1	1	1	1
		(24,24)	1	1	1	1	1
	4	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1
		(24,...,24)	1	1	1	1	1
	10	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1
		(24,...,24)	1	1	1	1	1

	100	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1
		(24,...,24)	1	1	1	1	1
	500	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1
		(24,...,24)	1	1	1	1	1
	1000	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1
		(24,...,24)	1	1	1	1	1
	10000	(2,...,2)	1	1	1	1	1
		(8,...,8)	1	1	1	1	1
		(16,...,16)	1	1	1	1	1

		(24,...,24)	1	1	1	1	1
Shallow	2	(2,2)	6	7	7	9	8
		(7,7)	11	10	11	14	12
		(14,14)	21	18	13	12	11
		(20,20)	12	15	11	16	19
	4	(2,...,2)	6	7	8	9	8
		(7,...,7)	11	10	11	14	12
		(14,...,14)	21	18	13	12	11
		(20,…,20)	15	15	11	16	19
	10	(2,...,2)	6	7	8	9	8
		(7,...,7)	11	11	11	14	12
		(14,...,14)	21	18	13	12	11
		(20,…,20)	15	15	12	16	19
	100	(2,...,2)	6	8	8	9	8
		(7,...,7)	11	11	11	14	12

		(14,...,14)	24	21	13	12	12
		(20,…,20)	18	18	12	18	19
	500	(2,...,2)	6	8	8	9	8
		(7,...,7)	11	11	11	14	12
		(14,...,14)	27	21	13	12	12
		(20,…,20)	18	18	12	18	21
	1000	(2,...,2)	6	8	8	9	8
		(7,...,7)	11	11	11	14	12
		(14,...,14)	27	21	13	14	12
		(20,…,20)	21	18	12	18	21
	10000	(2,...,2)	7	8	8	11	8
		(7,...,7)	11	11	11	15	14
		(14,...,14)	30	23	13	14	12
		(20,…,20)	23	21	12	18	21
Power	2	(2,2)	3	3	3	4	4

		(5,5)	3	3	3	4	4
		(7,7)	3	3	3	4	4
		(11,11)	3	3	3	4	4
	4	(2,...,2)	24	24	7	32	8
		(5,...,5)	26	25	7	34	8
		(7,...,7)	26	26	7	34	8
		(11,...,11)	27	27	7	36	9
	10	(2,...,2)	129	123	24	176	24
		(5,...,5)	133	131	24	187	24
		(7,...,7)	136	131	24	191	24
		(11,...,11)	139	136	24	195	25

APPENDIX 2

Numerical Result for CPU Time

Functions	Variables	Initial Points	TTRMIL	TTLAMR	TTKMAR	TTSMAR	TTSFA
Extended Rosenbrock	2	(2,2)	0.1095	0.1753	0.3185	0.0555	0.6094
		(5,5)	0.1038	0.0933	0.1053	0.0852	0.0693
		(7,7)	0.0717	0.0488	0.1062	0.0673	0.0605
		(9,9)	0.74	0.0876	0.1349	0.1063	0.043
	4	(2,...,2)	0.0842	0.0929	0.1491	0.109	0.0542
		(5,...,5)	0.066	0.093	0.0957	0.057	0.0533
		(7,...,7)	0.0507	0.1031	0.1118	0.0742	0.0567
		(9,...,9)	0.0959	0.0748	0.0999	0.0894	0.0754
	10	(2,...,2)	0.069	0.085	0.1511	0.0499	0.0647
		(5,...,5)	0.0606	0.0695	0.0894	0.0649	0.0443
		(7,...,7)	0.0711	0.0618	0.0964	0.0705	0.0626
		(9,...,9)	0.0647	0.0668	0.1011	0.0834	0.148

	100	(2,...,2)	0.1506	0.0933	0.1541	0.0367	0.0692
		(5,...,5)	0.0911	0.0872	0.0911	0.0718	0.0662
		(7,...,7)	0.0448	0.0497	0.1019	0.0503	0.0602
		(9,...,9)	0.0704	0.0537	0.088	0.0827	0.0673
	500	(2,...,2)	0.1226	0.097	0.1524	0.0449	0.0973
		(5,...,5)	0.0646	0.0747	0.0876	0.0621	0.0878
		(7,...,7)	0.0465	0.0498	0.0956	0.0581	0.1042
		(9,...,9)	0.0611	0.0956	0.1258	0.1023	0.0812
	1000	(2,...,2)	0.1471	0.1007	0.2027	0.0478	0.1835
		(5,...,5)	0.1087	0.0913	0.1173	0.0909	0.1156
		(7,...,7)	0.0683	0.1061	0.1405	0.0736	0.1673
		(9,...,9)	0.0678	0.1002	0.0824	0.098	0.1322
	10000	(2,...,2)	0.6523	0.4935	0.7799	0.2084	0.8325
		(5,...,5)	0.2606	0.308	0.5287	0.2954	0.7793
		(7,...,7)	0.2506	0.2376	0.4855	0.2858	0.7857

		(9,...,9)	0.2951	0.3741	0.4162	0.4964	0.8498
Extended White & Holst	2	(2,2)	0.065	0.0489	0.1288	0.0465	0.3318
		(5,5)	0.0523	0.0528	0.0818	0.0544	0.1053
		(9,9)	0.0866	0.0335	0.256	0.1117	0.1036
		(11,11)	0.0441	0.0523	0.1205	0.0748	0.1738
	4	(2,...,2)	0.0478	0.0462	0.1261	0.0462	0.1151
		(5,...,5)	0.0725	0.0585	0.0859	0.0575	0.0751
		(9,...,9)	0.0704	0.0437	0.288	0.1073	0.1065
		(11,...,11)	0.1073	0.0492	0.0562	0.1286	0.1395
	10	(2,...,2)	0.0445	0.056	0.1728	0.0583	0.1598
		(5,...,5)	0.0554	0.0637	0.0942	0.067	0.0724
		(9,...,9)	0.1015	0.041	0.2016	0.1612	0.102
		(11,...,11)	0.0506	0.0525	0.1864	0.1308	0.1739
	100	(2,...,2)	0.1078	0.106	0.3786	0.1392	0.2098
		(5,...,5)	0.1426	0.1417	0.265	0.1549	0.1832

		(9,...,9)	0.2727	0.0864	0.8779	0.2281	0.2313
		(11,...,11)	0.1227	0.1455	0.4905	0.322	0.5265
	500	(2,...,2)	0.4282	0.6214	1.5123	0.4751	0.5011
		(5,...,5)	0.7222	0.5924	1.0522	0.6167	0.5362
		(9,...,9)	0.9529	0.3655	2.2274	1.3288	0.7742
		(11,...,11)	0.4975	0.5306	1.2049	0.908	1.712
	1000	(2,...,2)	0.79	0.9286	2.5112	1.113	1.8661
		(5,...,5)	1.1713	1.0706	2.1765	1.0849	1.1109
		(9,...,9)	1.5477	0.7772	6.1708	1.5633	1.5342
		(11,...,11)	0.9399	1.1588	3.9457	1.9225	3.1269
Perturbed Quadratic	2	(3,3)	0.0597	0.0365	0.016	0.0082	0.0465
		(7,7)	0.0397	0.011	0.0096	0.0109	0.0194
		(9,9)	0.0093	0.01	0.0097	0.0117	0.0268
		(13,13)	0.0127	0.0061	0.0114	0.0118	0.014
	4	(3,...,3)	0.0264	0.0215	0.0115	0.0198	0.0231

		(7,...,7)	0.0502	0.039	0.0144	0.0345	0.0259
		(9,...,9)	0.0437	0.0707	0.0121	0.17	0.0232
		(13,...,13)	0.0328	0.0388	0.0156	0.0309	0.0195
	10	(3,...,3)	0.0649	0.0601	0.0311	0.0588	0.0321
		(7,...,7)	0.0453	0.0526	0.028	0.0602	0.0421
		(9,...,9)	0.0474	0.063	0.0284	0.059	0.0377
		(13,...,13)	0.0506	0.0548	0.022	0.0585	0.0359
Raydan 1	2	(3,3)	0.0808	0.0228	0.0584	0.0201	0.3347
		(8,8)	0.0189	0.0324	0.0129	0.0291	0.0439
		(13,13)	0.0114	0.0283	0.0156	0.0465	0.0472
		(18,18)	0.0453	0.0515	0.1241	0.0863	0.032
	4	(3,...,3)	0.0295	0.0437	0.0269	0.0262	0.0498
		(8,...,8)	0.0581	0.0408	0.0276	0.0417	0.0635
		(13,...,13)	0.049	0.0392	0.0486	0.0528	0.0579
		(18,...,18)	0.045	0.0594	0.0691	0.1402	0.0587

	10	(3,...,3)	0.0434	0.0734	0.0393	0.0509	0.0686
		(8,...,8)	0.1262	0.0865	0.053	0.0712	0.1351
		(13,...,13)	0.0662	0.066	0.0719	0.1705	0.1119
		(18,...,18)	0.0467	0.0448	0.0684	0.1022	0.1082
Extended Tridiagonal 1	2	(2,2)	0.1797	0.0661	0.0493	0.0441	0.0999
		(25,25)	0.0049	0.0065	0.0071	0.0519	0.0111
		(35,35)	0.004	0.0442	0.004	0.0057	0.0093
		(45,45)	0.0041	0.0043	0.0035	0.0041	0.0148
	4	(2,...,2)	0.1353	0.0609	0.0423	0.0362	0.0645
		(25,...,25)	0.0079	0.0077	0.004	0.0057	0.0117
		(35,...,35)	0.0056	0.0046	0.0042	0.0039	0.0061
		(45,...,45)	0.0052	0.0034	0.0041	0.0039	0.0083
	10	(2,...,2)	0.1857	0.0648	0.0724	0.0436	0.0661
		(25,...,25)	0.0055	0.0071	0.0094	0.0121	0.0177
		(35,...,35)	0.0039	0.0078	0.0051	0.0051	0.0157

		(45,...,45)	0.0044	0.0043	0.0049	0.0078	0.0459
	100	(2,...,2)	0.4582	0.1958	0.2599	0.3121	0.177
		(25,...,25)	0.12	0.0731	0.6522	0.0477	0.0689
		(35,...,35)	0.0148	0.0149	0.0292	0.0137	0.0233
		(45,...,45)	0.014	0.0136	0.0132	0.016	0.0286
	500	(2,...,2)	1.5569	0.7726	1.1597	1.1079	0.4211
		(25,...,25)	0.555	0.2756	2.3461	0.6834	0.2251
		(35,...,35)	0.0437	0.0468	0.0441	0.0438	0.073
		(45,...,45)	0.0456	0.0435	0.0434	0.0446	0.0689
	1000	(2,...,2)	4.201	2.7154	2.2885	3.7762	0.8265
		(25,...,25)	1.4311	0.7726	4.5908	2.2244	0.8265
		(35,...,35)	0.0824	0.0795	0.1161	0.0811	0.1388
		(45,...,45)	0.0814	0.0806	0.1079	0.0806	0.1366
Diagonal 4	2	(5,5)	0.0088	0.0088	0.0036	0.0083	0.0321
		(10,10)	0.0069	0.005	0.0036	0.0108	0.0182

		(20,20)	0.0043	0.005	0.0068	0.0079	0.016
		(30,30)	0.0086	0.0053	0.0059	0.0086	0.0378
	4	(5,...,5)	0.007	0.0053	0.0078	0.0167	0.0155
		(10,...,10)	0.0085	0.0047	0.0048	0.0106	0.0192
		(20,...,20)	0.0093	0.0106	0.0093	0.0131	0.0277
		(30,...,30)	0.0106	0.0091	0.0139	0.0107	0.0187
	10	(5,...,5)	0.0071	0.0088	0.0091	0.012	0.0205
		(10,...,10)	0.0093	0.0114	0.0073	0.0093	0.0187
		(20,...,20)	0.0131	0.0091	0.0323	0.0117	0.0263
		(30,...,30)	0.0086	0.0111	0.0129	0.0132	0.0221
	100	(5,...,5)	0.017	0.0108	0.0095	0.0107	0.0337
		(10,...,10)	0.0095	0.0108	0.0067	0.0104	0.0227
		(20,...,20)	0.0103	0.0093	0.0129	0.0145	0.0503
		(30,...,30)	0.0753	0.0208	0.0117	0.0214	0.0533
	500	(5,...,5)	0.0098	0.0145	0.0136	0.0144	0.0328

		(10,...,10)	0.0169	0.0108	0.0086	0.0148	0.0557
		(20,...,20)	0.0153	0.0146	0.0092	0.0158	0.0455
		(30,...,30)	0.0106	0.0092	0.0106	0.0148	0.0356
	1000	(5,...,5)	0.0138	0.0168	0.0128	0.0171	0.0454
		(10,...,10)	0.0134	0.0119	0.0158	0.0175	0.0569
		(20,...,20)	0.0124	0.0111	0.0116	0.0273	0.0683
		(30,...,30)	0.0132	0.0122	0.0121	0.0192	0.0578
	10000	(5,...,5)	0.0555	0.0398	0.0403	0.0622	0.3475
		(10,...,10)	0.0424	0.0404	0.1225	0.066	0.3325
		(20,...,20)	0.0384	0.0411	0.0437	0.0724	0.5101
		(30,...,30)	0.0471	0.0502	0.0429	0.0737	0.4721
Extended Himmelblau	2	(2,2)	0.0381	0.0254	0.0156	0.0162	0.0471
		(6,6)	0.0161	0.0143	0.014	0.0279	0.042
		(14,14)	0.0272	0.0178	0.0172	0.0219	0.0394
		(24,24)	0.0132	0.0135	0.013	0.0213	0.0488

	4	(2,...,2)	0.0167	0.0167	0.0103	0.0148	0.0388
		(6,...,6)	0.0182	0.0204	0.0105	0.0215	0.0368
		(14,...,14)	0.0212	0.0147	0.0129	0.0212	0.0382
		(24,…,24)	0.0149	0.0205	0.0134	0.0269	0.0374
	10	(2,...,2)	0.0141	0.0145	0.0118	0.0121	0.0449
		(6,...,6)	0.0197	0.0147	0.0133	0.0277	0.051
		(14,...,14)	0.0229	0.0187	0.0116	0.0302	0.0416
		(24,…,24)	0.0202	0.0236	0.0152	0.0243	0.0425
	100	(2,...,2)	0.0203	0.0186	0.0138	0.0166	0.0372
		(6,...,6)	0.021	0.023	0.015	0.0317	0.0528
		(14,...,14)	0.023	0.024	0.0191	0.0302	0.0428
		(24,…,24)	0.0223	0.0213	0.0153	0.036	0.0623
	500	(2,...,2)	0.0373	0.0332	0.0234	0.035	0.0604
		(6,...,6)	0.0367	0.0374	0.0203	0.0504	0.0952
		(14,...,14)	0.0306	0.0312	0.0316	0.0584	0.0598

		(24,…,24)	0.0287	0.0284	0.0228	0.0478	0.0615
	1000	(2,...,2)	0.0417	0.0405	0.0304	0.0344	0.1124
		(6,...,6)	0.0916	0.0524	0.0262	0.0773	0.1591
		(14,...,14)	0.051	0.0552	0.0288	0.0651	0.0991
		(24,…,24)	0.041	0.0402	0.0381	0.0644	0.0969
	10000	(2,...,2)	0.1179	0.1059	0.0789	0.1477	0.527
		(6,...,6)	0.1176	0.1239	0.0807	0.1785	0.5426
		(14,...,14)	0.1208	0.1139	0.0789	0.1817	0.6231
		(24,…,24)	0.1066	0.1081	0.0746	0.1617	0.5302
Extended DENSCHNB	2	(3,3)	0.0266	0.0108	0.0123	0.0129	0.0444
		(8,8)	0.0149	0.0186	0.0175	0.0144	0.0335
		(15,15)	0.0149	0.0152	0.0153	0.0389	0.0381
		(21,21)	0.0203	0.0158	0.019	0.0273	0.0377
	4	(3,...,3)	0.0151	0.0169	0.0154	0.0115	0.0356
		(8,...,8)	0.0247	0.023	0.0207	0.0139	0.0314

		(15,...,15)	0.0139	0.0208	0.0135	0.0278	0.0462
		(21,…,21)	0.0151	0.0235	0.0128	0.0341	0.0395
	10	(3,...,3)	0.013	0.0123	0.0132	0.0125	0.0292
		(8,...,8)	0.0167	0.0182	0.0158	0.016	0.0297
		(15,...,15)	0.0144	0.0172	0.0201	0.0326	0.0473
		(21,…,21)	0.0196	0.015	0.0138	0.0311	0.0391
	100	(3,...,3)	0.0175	0.0156	0.0127	0.0159	0.0496
		(8,...,8)	0.016	0.0232	0.0195	0.028	0.0496
		(15,...,15)	0.0183	0.0204	0.0153	0.0373	0.0446
		(21,…,21)	0.0197	0.0243	0.0184	0.0367	0.0571
	500	(3,...,3)	0.0506	0.0253	0.0205	0.0284	0.0496
		(8,...,8)	0.0258	0.0308	0.0275	0.0292	0.0574
		(15,...,15)	0.031	0.0304	0.0305	0.0541	0.0778
		(21,…,21)	0.0313	0.048	0.0296	0.1147	0.0756
	1000	(3,...,3)	0.0355	0.0291	0.0257	0.0319	0.0803

		(8,...,8)	0.044	0.0425	0.0607	0.0419	0.1089
		(15,...,15)	0.0363	0.0409	0.0376	0.1128	0.1047
		(21,…,21)	0.0459	0.0383	0.0417	0.0762	0.0905
	10000	(3,...,3)	0.0979	0.0746	0.0743	0.0903	0.4744
		(8,...,8)	0.1304	0.1048	0.1047	0.1108	0.3869
		(15,...,15)	0.1057	0.1211	0.104	0.2074	0.5181
		(21,…,21)	0.0935	0.1136	0.1014	0.2701	0.4899
FLETCHCR	2	(5,5)	0.0672	0.0223	0.0221	0.0126	0.0389
		(10,10)	0.0154	0.0116	0.0096	0.0127	0.0294
		(15,15)	0.0116	0.0086	0.01	0.0115	0.0123
		(20,20)	0.009	0.0069	0.0106	0.0104	0.0138
	4	(5,...,5)	0.035	0.0319	0.0257	0.0274	0.0562
		(10,...,10)	0.0363	0.0322	0.0359	0.0406	0.041
		(15,...,15)	0.0559	0.0457	0.0284	0.0486	0.0452
		(20,…,20)	0.0418	0.0328	0.0346	0.0439	0.0399

	10	(5,...,5)	0.0422	0.052	0.0437	0.049	0.0706
		(10,...,10)	0.0516	0.0522	0.0459	0.0619	0.0908
		(15,...,15)	0.066	0.066	0.08	0.08	0.1798
		(20,…,20)	0.0694	0.0763	0.0735	0.1042	0.1084
Hager	2	(4,4)	0.0119	0.0119	0.0121	0.0159	0.1557
		(8,8)	0.0132	0.0146	0.0095	0.0154	0.053
		(18,18)	0.0134	0.0149	0.0161	0.0094	0.0819
		(21,21)	0.0137	0.0239	0.0172	0.0413	0.0246
	4	(4,...,4)	0.0228	0.0192	0.0197	0.0159	0.0251
		(8,...,8)	0.0199	0.0173	0.0182	0.0185	0.0606
		(18,...,18)	0.0197	0.0275	0.0204	0.0173	0.0333
		(21,…,21)	0.4061	0.392	0.3297	0.2933	0.059
	10	(4,...,4)	0.0274	0.0252	0.0221	0.0737	0.0287
		(8,...,8)	0.0209	0.0219	0.0213	0.0222	0.0282
		(18,...,18)	0.0316	0.0251	0.024	0.0218	0.0805

		(21,…,21)	0.079	0.0808	0.0888	0.0605	0.0805
	100	(4,...,4)	0.0919	0.0754	0.0757	0.1441	0.0858
		(8,...,8)	0.0643	0.1404	0.0766	0.0703	0.0818
		(18,...,18)	0.1514	0.0949	0.0745	0.086	0.0966
		(21,…,21)	0.4626	0.3931	0.2903	0.3533	0.2437
Extended Penalty	2	(3,3)	0.0524	0.0391	0.0249	0.0275	0.1099
		(7,7)	0.0268	0.0153	0.0221	0.024	0.0206
		(10,10)	0.0218	0.0249	0.0192	0.0258	0.0183
		(23,23)	0.0155	0.0135	0.0191	0.0179	0.0306
	4	(3,...,3)	0.0397	0.0291	0.0112	0.0276	0.0992
		(7,...,7)	0.0328	0.0353	0.0142	0.0265	0.0234
		(10,...,10)	0.0483	0.0266	0.0203	0.0246	0.0304
		(23,...,23)	0.0138	0.0181	0.0148	0.0237	0.0208
	10	(3,...,3)	0.0483	0.0498	0.0186	0.0274	0.0204
		(7,...,7)	0.0406	0.0333	0.0168	0.0176	0.0403

		(10,...,10)	0.024	0.0264	0.0153	0.0253	0.0202
		(23,...,23)	0.041	0.0322	0.015	0.0203	0.0317
	100	(3,...,3)	0.0893	0.0368	0.03	0.0676	FAIL
		(7,...,7)	FAIL	0.1555	FAIL	0.045	FAIL
		(10,...,10)	0.0383	0.0292	FAIL	0.0256	0.3254
		(23,...,23)	0.0338	0.0479	FAIL	0.0248	0.1095
Extended Maratos	2	(2,2)	0.0731	0.0359	0.0481	0.0318	0.1349
		(5,5)	0.0388	0.0318	0.0557	0.0556	0.1829
		(8,8)	0.0434	0.0389	0.0441	0.044	0.1261
		(11,11)	0.0377	0.0424	0.0318	0.041	0.1389
	4	(2,...,2)	0.0369	0.0293	0.0446	0.0287	0.0834
		(5,...,5)	0.0363	0.0333	0.0494	0.0405	0.1732
		(8,...,8)	0.0433	0.0333	0.0778	0.0398	0.1566
		(11,...,11)	0.0383	0.0448	0.0334	0.0473	0.1265
	10	(2,...,2)	0.0346	0.0486	0.0793	0.0379	0.0493

		(5,...,5)	0.0356	0.0455	0.0529	0.0391	0.1155
		(8,...,8)	0.0509	0.055	0.0342	0.0494	0.1608
		(11,...,11)	0.0339	0.0487	0.0449	0.0761	0.1108
Sphere	2	(2,2)	0.0093	0.0049	0.0056	0.0029	0.0087
		(8,8)	0.0034	0.0039	0.0038	0.0041	0.0077
		(16,16)	0.0063	0.0037	0.0029	0.0036	0.0057
		(24,24)	0.0038	0.0025	0.0026	0.004	0.0059
	4	(2,...,2)	0.0041	0.0026	0.0036	0.0039	0.0109
		(8,...,8)	0.003	0.0036	0.0047	0.0028	0.004
		(16,...,16)	0.0052	0.0047	0.0046	0.0032	0.0059
		(24,...,24)	0.0056	0.0028	0.0155	0.006	0.0039
	10	(2,...,2)	0.0049	0.004	0.0056	0.006	0.0037
		(8,...,8)	0.0116	0.0167	0.0034	0.0068	0.0038
		(16,...,16)	0.0036	0.0041	0.0061	0.0033	0.0065
		(24,...,24)	0.0073	0.0038	0.0038	0.0029	0.0049

	100	(2,...,2)	0.0031	0.0032	0.0056	0.0027	0.0051
		(8,...,8)	0.0052	0.003	0.0044	0.005	0.0078
		(16,...,16)	0.003	0.0044	0.0067	0.0058	0.0485
		(24,...,24)	0.007	0.0053	0.0029	0.0058	0.0061
	500	(2,...,2)	0.0043	0.0033	0.0052	0.0043	0.0107
		(8,...,8)	0.0032	0.0041	0.0054	0.0052	0.0064
		(16,...,16)	0.004	0.0046	0.0042	0.0034	0.005
		(24,...,24)	0.0035	0.0047	0.0045	0.0051	0.0048
	1000	(2,...,2)	0.0038	0.0044	0.0055	0.0043	0.0081
		(8,...,8)	0.004	0.0066	0.0067	0.0036	0.0091
		(16,...,16)	0.0051	0.0046	0.0043	0.0036	0.0057
		(24,...,24)	0.0037	0.0045	0.0045	0.0038	0.0056
	10000	(2,...,2)	0.0236	0.0464	0.0107	0.0089	0.0545
		(8,...,8)	0.0099	0.0106	0.0091	0.0086	0.0177
		(16,...,16)	0.0103	0.0128	0.0094	0.0093	0.0197

		(24,...,24)	0.0133	0.0126	0.0115	0.0102	0.0245
Shallow	2	(2,2)	0.045	0.022	0.0154	0.0208	0.1197
		(7,7)	0.019	0.0176	0.0193	0.0301	0.1263
		(14,14)	0.0396	0.0415	0.0211	0.0225	0.0492
		(20,20)	0.0294	0.0264	0.0231	0.0266	0.1662
	4	(2,...,2)	0.0139	0.0126	0.0158	0.0264	0.0313
		(7,...,7)	0.0583	0.0207	0.0255	0.0303	0.0291
		(14,...,14)	0.0354	0.0304	0.0223	0.0241	0.0291
		(20,…,20)	0.0228	0.052	0.0172	0.0309	0.0326
	10	(2,...,2)	0.0122	0.0139	0.0149	0.0168	0.0186
		(7,...,7)	0.0251	0.0218	0.0204	0.0272	0.0522
		(14,...,14)	0.0372	0.0357	0.0222	0.0251	0.0223
		(20,…,20)	0.0339	0.0259	0.0213	0.0313	0.034
	100	(2,...,2)	0.021	0.0243	0.0255	0.0431	0.021
		(7,...,7)	0.0397	0.0291	0.0506	0.0307	0.0328

		(14,...,14)	0.1099	0.0424	0.0285	0.0362	0.0264
		(20,…,20)	0.0462	0.048	0.0241	0.0356	0.0399
	500	(2,...,2)	0.0192	0.0262	0.0252	0.0283	0.05
		(7,...,7)	0.0382	0.0354	0.0746	0.0504	0.0507
		(14,...,14)	0.0858	0.0708	0.0502	0.0738	0.0407
		(20,…,20)	0.0606	0.0562	0.0461	0.0599	0.1364
	1000	(2,...,2)	0.0301	0.0545	0.0619	0.0427	0.0427
		(7,...,7)	0.0515	0.0505	0.0495	0.0863	0.0515
		(14,...,14)	0.1404	0.1029	0.0668	0.1186	0.0524
		(20,…,20)	0.1729	0.0893	0.0539	0.0934	0.0909
	10000	(2,...,2)	0.0959	0.0838	0.0824	0.1224	0.2666
		(7,...,7)	0.1126	0.1116	0.112	0.1638	0.346
		(14,...,14)	0.3643	0.2379	0.1325	0.1555	0.3125
		(20,…,20)	0.233	0.2217	0.1254	0.186	0.5557
Power	2	(2,2)	0.0084	0.0105	0.0085	0.0109	0.023

		(5,5)	0.0064	0.0057	0.009	0.0097	0.0151
		(7,7)	0.0077	0.0064	0.0094	0.0081	0.0151
		(11,11)	0.0069	0.0072	0.0069	0.0096	0.0151
	4	(2,...,2)	0.0366	0.0408	0.0147	0.0564	0.0334
		(5,...,5)	0.059	0.0401	0.0113	0.07	0.0193
		(7,...,7)	0.0584	0.0445	0.0144	0.0543	0.0188
		(11,...,11)	0.0474	0.0488	0.0151	0.0701	0.018
	10	(2,...,2)	0.2378	0.2117	0.0375	0.2853	0.0538
		(5,...,5)	0.2336	0.2206	0.0543	0.3283	0.0537
		(7,...,7)	0.2192	0.3215	0.0542	0.3408	0.0511
		(11,...,11)	0.183	0.206	0.0473	0.3488	0.055

APPENDIX 3

MATLAB Code Programming for Three-Term Conjugate Gradient Methods

```
function [x,k,t]=CG(x0,OP1,OP2)
clear t;
tic
%This function for finding the minimizer of
%a given function with initial point x0
%OP1 for the type of Beta
%and OP2 for the line search
%and OP3 for the testing function
g=test_function(x0,OP2,1);
d=-g;
k=0;
x=x0;
while norm(g)>.000001
    Alpha=L_search(x,d,OP2);
    t=g;
    x=x+Alpha*d;
    g=test_functions(x,OP2,1);

    switch OP1

         case 'RMIL'
             Beta = (g'*(g-t))/(d'*d);
        case 'SMAR'
            Beta = (g'*(g-((norm(g)/norm(t))*d)))/(d'*d);
        case 'KMAR'
            Beta = (g'*(g-t))/(t'*(g+t));
        case 'LAMR'
            Beta = (g'*(((norm(d)/norm(d-g))*g)-t))/((norm(d)/norm(d-g))*(norm(d)^2));
        case 'SFA'
            Beta = -(g'*(g-((norm(g)/norm(t))*t)))/(d'*t);

        otherwise
    error('Unknown Beta')
    end

    d=-g+Beta*d-(g-t)*((g'*d)/(d'*(d-g)));
    k=k+1;
end
t=toc;
```

APPENDIX 4

MATLAB Code Programming for Test Functions

```
function f=test_functions(x,fn,index)

n = length(x);
if nargin <3
index=0;
end
switch fn

    case 1 %Extended Rosenbrock
        if index == 0
            f=0;
            for j=1:n/2
            f=f+100*(x(2*j)-x(2*j-1)^2)^2+(1-x(2*j-1))^2;
            end
        elseif index ==1
            for j=1:n/2
                h(2*j-1)=-400*(x(2*j)-x(2*j-1)^2)*x(2*j-1)-2+2*x(2*j-1);
                h(2*j)=200*x(2*j)-200*(x(2*j-1)).^2;
            end
            f=h';
        else
            error('Index must be a boolean')
        end


    case 2 %Ext. White & Holst
        if index == 0
            f=0;
            for i=1:n/2
            f = f + 100*(x(2*i)-x(2*i-1)^3)^2 + (1-x(2*i-1))^2;
            end
        elseif index ==1
            j=1;
            for i=1:n/2
                h(j) = -6*100*x(2*i-1)*x(2*i-1)*(x(2*i)-x(2*i-1)^3)-2*(1-x(2*i-1));
                h(j+1) =  2*100*(x(2*i)-x(2*i-1)^3);
                j = j + 2;
            end
            f=h';
        else
            error('Index must be a boolean')
        end
```

```
case 3 %Perturbed Quadratic
    if index ==0
        f=0;s=0;t=0;
        for j=1:n
            s=s+j*(x(j)^2);
            t=t+x(j);
        end
        f=s+(1/100)*t^2;
    elseif index ==1
        s=0;
        for j=1:n
            s=s+x(j);
        end
        for i=1:n
        h(i)=2*i*x(i)+(2/100)*s;
        end
        f=h';
    else
        error('Index must be a boolean')
    end

case 4 %Raydan 1
    if index ==0
        f=0;
        for j=1:n
            f=f+(j/10)*(exp(x(j))-x(j));
        end
    elseif index ==1
        for j=1:n
           h(j)=(j/10)*(exp(x(j))-1);
        end
        f=h';
    else
        error('Index must be a boolean')
    end

case 5 %Ext. Tridiagonal 1
    if index ==0
        f=0;
        for j=1:n/2
            f=f+(x(2*j-1)+x(2*j)-3)^2+(x(2*j-1)-x(2*j)+1)^4;
        end
    elseif index ==1
        for j=1:n/2
            h(2*j-1)=2*(x(2*j-1)+x(2*j)-3)+4*(x(2*j-1)-x(2*j)+1)^3;
            h(2*j)=2*(x(2*j-1)+x(2*j)-3)-4*(x(2*j-1)-x(2*j)+1)^3;
        end
        f=h';
    else
        error('Index must be a boolean')
    end
```

```
case 6 %Diagonal 4
    if index ==0
        f=0;
        for j=1:n/2
            f=f+(1/2)*((x(2*j-1)^2)+(100*x(2*j)^2));
        end
    elseif index ==1
        for j=1:n/2
            h(2*j-1)=x(2*j-1);
            h(2*j)=100*x(2*j);
        end
        f=h';
    else
        error('Index must be a boolean')
    end
```

```
case 7 %Ext. Himmelblau
    if index ==0
        f=0;
        for j=1:n/2
            f=f+(x(2*j-1)^2+x(2*j)-11)^2+(x(2*j-1)+x(2*j)^2-7)^2;
        end
    elseif index ==1
        for j=1:n/2
            h(2*j-1)=2*(x(2*j-1)^2+x(2*j)-11)*(2*x(2*j-1))+2*(x(2*j-1)+x(2*j)^2-7);
            h(2*j)=2*(x(2*j-1)^2+x(2*j)-11)+2*(x(2*j-1)+x(2*j)^2-7)*(2*x(2*j));
        end
        f=h';
    else
        error('Index must be a boolean')
    end
```

```
case 8 %Ext. DENSCHNB
    if index ==0
        f=0;
        for i=1:n/2
            f=f+(x(2*i-1)-2)^2+(x(2*i-1)-2)^2*x(2*i)^2+(x(2*i)+1)^2;
        end
    elseif index ==1
        for i=1:n/2
            h(2*i-1)=2*(x(2*i-1)-2)+2*(x(2*i-1)-2)*(x(2*i)^2);
            h(2*i)=2*(x(2*i-1)-2)^2*x(2*i)+2*(x(2*i)+1);
        end
        f=h';
    else
        error('Index must be a boolean')
    end
```

```
case 9 %FLETCHCR
    if index ==0
        f=0;
        for j=1:n-1
            f=f+100*(x(j+1)-x(j)+1-x(j)^2)^2;
        end
    elseif index ==1
        h(1)=200*(x(2)-x(1)+1-x(1)^2)*(-1-2*(x(1)));
        for j=2:n-1
            h(j)=200*(x(j)-x(j-1)+1-x(j-1)^2)+200*(x(j+1)-x(j)+1-x(j)^2)*(-1-2*x(j));
        end
        h(n)=200*(x(n)-x(n-1)+1-x(n-1)^2);
        f=h';
    else
        error('Index must be a boolean')
    end

    case 10 %Hager
        if index ==0
            f=0;
            for i=1:n
            f=f+(exp(x(i))-sqrt(i)*x(i));
            end
        elseif index ==1
            for i=1:n
                h(i)=(exp(x(i))-sqrt(i));
            end
            f=h';
        else
            error('Index must be a boolean')
        end

    case 11 %Extended Penalty
        if index ==0
            f=0;s=0;t=0;

            for j=1:n
                s=s+(x(j)^2-0.25);
            end
            for j=1:n-1
                t=t+(x(j)-1)^2;
            end
            f=t+s^2;

        elseif index ==1
            s=0;
            for j=1:n
                s=s+(x(j)^2-0.25);
            end
            for i=1:n-1
            h(i)=2*(x(i)-1)+2*s*2*x(i);
            end
            h(n)=2*s*2*x(n);
            f=h';
        else
            error('Index must be a boolean')
        end
```

```
case 12 %Extended Maratos
    if index ==0
        f=0;
        for j=1:n/2
            f=f+x(2*j-1)+100*(x(2*j-1)^2+x(2*j)^2-1)^2;
        end
    elseif index ==1
        for j=1:n/2
            h(2*j-1)=1+200*(x(2*j-1)^2+x(2*j)^2-1)*2*x(2*j-1);
            h(2*j)=200*(x(2*j-1)^2+x(2*j)^2-1)*2*x(2*j);
        end
        f=h';
    else
        error('Index must be a boolean')
    end

case 13 %Sphere
    if index ==0
        f = 0;
        for i = 1:n
            f = f+x(i)^2;
        end
    elseif index ==1
        for i=1:n
            h(i)=2*x(i);
        end
        f=h';
    else
        error('Index must be a boolean')
    end

case 14 %shallow
        if index ==0
        f=0;
        for j=1:n/2
            f=f+(x(2*j-1)^2-x(2*j))^2+(1-x(2*j-1))^2;
        end
        elseif index ==1
            for j=1:n/2
                h(2*j-1)=2*(x(2*j-1)^2-x(2*j))*2*x(2*j-1)-2*(1-x(2*j-1));
                h(2*j)=-2*(x(2*j-1)^2-x(2*j));
            end
            f=h';
        else
            error('Index must be a boolean')
        end
```

```
    case 15 %Power
        if index ==0
            f=0;
            for i=1:n
                f=f+(i*x(i))^2;
            end
        elseif index ==1
            for i=1:n
                h(i)=2*i^2*x(i);
                %h(2*i)=2*x(2*i-1)*(x(2*i-1)*x(2*i)-50000);
            end
            f=h';
        else
            error('Index must be a boolean')
        end
end
```

APPENDIX 5

MATLAB Code Programming for IMRT Fluence Map Optimization

```
% Setting up problem with FluenceMapOpt
clear all; close all; clc;

% PTV - prostate
ptv.name = 'PTV_68';
ptv.terms = {struct('type','unif','dose',81,'weight',1)};

% OAR - rectum
oar.name = 'Rectum';
oar.terms = {struct('type','udvc','dose',50,'percent',50,'weight',1)};

% Create problem instance
structs = {ptv,oar};
prob = FluenceMapOpt(structs);

% Calculate beamlet intensities
prob.calcBeams();

% Plot objective function
prob.plotObj();

% Plot dose-volume histogram
prob.plotDVH();

% Plot beamlet intensities
prob.plotBeams();

% Plot dose
prob.plotDose();
```